

Inverse kinematics of 7-DOF multi-offset manipulators for on-orbit servicing based on variable SRS equivalence and hybrid optimization

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Abstract

The 7-DOF manipulator with multi-offset offers higher operational flexibility and a larger workspace, widely used in space on-orbit servicing. This paper proposes a novel inverse kinematics solution for 7-DOF multi-offset manipulators based on the concept of variable SRS equivalent manipulators and arm angle. First, a new definition scheme for shoulder and wrist points is proposed to address failures in traditional methods. Then, the concept of the variable SRS equivalent manipulator is proposed, solving the problem of existing arm angle based methods being inapplicable to the manipulator configuration studied in this paper. Subsequently, integrating this concept with arm angle method, the complex problem of solving seven joint angles within joint limits is transformed into solving a system of two nonlinear equations with two unknowns in the feasible domain. To address this nonlinear system, a hybrid algorithm combining an improved Levenberg–Marquardt method and an improved Particle Swarm Optimization algorithm is developed, augmented with an efficient initial value estimation strategy. The proposed inverse kinematics solution successfully introduces arm angle as a redundant parameter, filling the research gap in applying arm angle-based solutions to the studied manipulator configuration. Finally, verification and comparative experiments show that our inverse kinematics solution achieves higher success rates, efficiency, and safety in solving. © 2026 COSPAR. Published by Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Keywords: Redundant manipulators; Offset manipulator; Inverse kinematics; Arm angle; Nonlinear equations

1. Introduction

The rapid advancement of space exploration programs and space-based infrastructure construction has established modular assembly of ultra-large orbital facilities as a pivotal challenge in space technology development (Flores-

Abad et al., 2014). Current payload limitations of launch vehicles render monolithic spacecraft incapable of directly deploying megastructures such as kilometer-scale space telescopes and multi-module space stations (Li et al., 2019). To address this constraint, on-orbit servicing systems have pioneered an innovative “modular launch and orbital assembly” paradigm, with space robotic manipulators serving as pivotal enablers (Zhao et al., 2025). Notably, the European Space Agency (ESA) has developed a novel multi-arm reconfigurable robotic system under its

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“MIRROR” project (Deremetz et al., 2023). This system employs a 7-degree-of-freedom (7-DOF) robotic manipulator featuring multiple joint offsets, which confers a symmetric joint configuration that allows bidirectional installation without structural modifications. The multi-offset architecture enhances operational flexibility and workspace coverage, making it particularly suitable for complex extraterrestrial missions (Deremetz et al., 2022). Comparable kinematic configurations have been implemented in other robotic systems, exemplified by the YuMi collaborative robot by ABB (Asgari et al., 2025). However, the redundant 7-DOF architecture with shoulder-elbow-wrist offsets introduces significant complexities in inverse kinematics (IK) resolution. Furthermore, stringent requirements are imposed on solving precision, computational efficiency, and operational safety for space applications (Zhao et al., 2025).

This study focuses on the inverse kinematics solution for a 7-DOF manipulator with multi-offset features at the shoulder, elbow, and wrist. Its research motivation stems directly from the urgent needs of on-orbit servicing for future large-scale space infrastructures. As mentioned in the relevant literature, payload fairing size limitations of launch vehicles necessitate the adoption of a “modular launch and on-orbit assembly” approach for ultra-large structures such as kilometer-scale telescopes (Deremetz et al., 2022; Deremetz et al., 2023). Under this paradigm, relocatable space manipulators have become core enabling technologies for achieving autonomous on-orbit assembly. The configuration studied in this paper, similar to advanced systems like the ESA “MIRROR” project and the ABB YuMi robot, features multi-offset and symmetric designs that endow the manipulator with bidirectional installation capability and an expanded workspace. This is crucial for locomotion and precision manipulation in confined, complex on-orbit environments. For instance, this configuration enables the manipulator to “walk” on large structures and accurately transport modules (Flores-Abad et al., 2014; Deremetz et al., 2023; Zhao et al., 2025). Therefore, overcoming the challenges of efficient and stable inverse kinematics solutions for such complex configurations is a key prerequisite for enhancing the autonomy, reliability, and cost-effectiveness of on-orbit assembly tasks, and it represents the core value of this research.

Current research on inverse kinematics solutions for manipulators with offsets has yielded several methodological advancements, though notable limitations persist. Wu et al. (2013) simplified wrist-offset manipulators into non-offset virtual arms. They utilized the inverse kinematic solutions of the virtual arm as initial approximations and iteratively approached the actual solutions. Nevertheless, their fixed virtual wrist center assumption has limits. Kucuk and Bingul (2014) developed the Numerical Inverse Kinematics Algorithm (NIKA), demonstrating efficacy for wrist-offset manipulators, though its robustness requires further validation in generalized scenarios. Yang and

Zhang (2018) proposed a hybrid geometric-algebraic method incorporating elbow angles as redundant parameters to derive analytical solutions, yet its applicability is constrained by manipulator-specific geometric structures. Sinha and Chakraborty (2019) established an IK resolution framework grounded in geometric search-space exploration principles, but it has limitations in computational efficiency for real-time applications due to the discretization process involved. Li et al. (2021a) compensated for end-effector positional deviations through structural mapping between simplified and original manipulators, but the error is relatively large. Ho et al. (2023) implemented deep learning for redundant inverse kinematics, though joint solution accuracy remains inferior to numerical methods.

The IK challenge for 7-DOF manipulators with shoulder, wrist, and elbow offsets is more difficult. Luo et al. (2014) developed closed-form solutions via fixed-joint parameterization, limited by computational efficiency and redundancy selection constraints. Zaplana et al. (2018) combined analytical and numerical methods for ABB YuMi, constructing variant robots compliant with Pieper’s criteria to initialize solutions, yet stability and speed limitations persist. Alebooyeh and Urbanic (2019) employed artificial neural networks (ANN) for ABB YuMi, facing challenges in precision, generalization, and real-time performance. Dereli and Koker (2020) applied firefly algorithms to 7-DOF systems, susceptible to local optima in complex configuration spaces. Yang et al. (2024) presented a novel IK solver for a 7-DOF manipulator with offset, combining gradient and geometric approaches to efficiently solve joint angles while considering constraints, but it has limited experimental validation.

Furthermore, several general-purpose inverse kinematics methods demonstrate applicability to 7-DOF manipulators with multiple joint offsets. The Kinematics and Dynamics Library (KDL) solver (Newman, 2002) employs a numerical approach based on Jacobian matrix iterations and optimization algorithms, incrementally refining joint angles to minimize pose errors until converging to valid solutions. The TRAC-IK solver (Beeson and Ames, 2015) enhances the conventional KDL framework through a dual-algorithm parallel architecture. Its core innovation lies in integrating an improved KDL stochastic restart module with a sequential quadratic programming (SQP) optimizer, achieving significant improvements in solution search efficiency and success rates. Bio IK (Starke, 2020) introduces a multi-algorithm fusion strategy, combining genetic algorithm selection-crossover mechanisms with particle swarm optimization for global exploration, while incorporating L-BFGS-B for local refinement. This hierarchical architecture enhances global optimization capability. However, such hybrid methods may induce abrupt joint trajectory discontinuities during convergence. For spaceborne manipulators executing precision orbital operations, these discontinuities pose critical safety risks.

Arm angle parameterization effectively characterizes manipulator configurations, and inverse kinematics methods based on this approach exhibit unique advantages in obstacle avoidance and configuration optimization for 7-DOF manipulators (Oh et al., 2017; Yan et al., 2020; Xu et al., 2024). Research on arm angle-based methods remains an active area of investigation. Kreutzdelgado et al. (1992) pioneered the arm angle parameterization model by establishing geometric mappings among shoulder-elbow-wrist joints, transforming redundant DOFs into quantifiable parameters to enhance configuration space controllability. While this method laid the theoretical foundation for subsequent studies, it lacked robust singularity avoidance mechanisms. Shimizu et al. (2008) developed a dynamic reference plane selection algorithm combined with joint range constraints, reducing singularity occurrence rates. However, multi-solution reference planes introduced significant end-effector orientation deviations, limiting precision assembly applications. Zhou et al. (2015) incorporated joint limits and dynamic stability constraints into geometric parameterization frameworks, proposing an arm angle screening strategy based on feasible solution sets, albeit with computational inefficiency. Xu et al. (2016) developed a dual-wrist joint coupling model with compensation coefficients to resolve closed-chain anomalies, though underdetermined correlation equations induced multi-manipulator coordination errors. Jin et al. (2020) proposed a hybrid optimization model based on elbow pointing angle and Particle Swarm Optimization (PSO) enabling obstacle avoidance, yet demonstrated limited generalizability and stability. Ma et al. (2022) enhanced end-effector positioning accuracy for SSRMS manipulators through pseudo-inverse Jacobian matrix corrections with dynamic damping coefficients, though requiring extended computation time. Li et al. (2022) presented an IK method, featuring an analytical IK solution and a natural arm configuration learning method for human-like motion, but it has limited generalization to different robot structures. Zhao et al. (2023) introduced a multi-objective optimization framework for SSRMS manipulators based on arm angle parameterization, achieving exceptional computational efficiency and solution accuracy suitable for space station payload operations. However, the versatility is not strong. Elias and Wen (2024) generalized Shoulder-Elbow-Wrist angle method for 7-DOF manipulators reduces singularities and expands workspace utility, but still faces computational complexity in specific robot configurations.

In summary, arm-angle parameterization methods demonstrate unique value in solving the inverse kinematics of 7-DOF manipulators. By introducing the arm angle ψ as a redundant parameter, this approach offers significant advantages for online applications: in obstacle avoidance planning, the arm angle serves as an effective control variable to optimize the manipulator's configuration and prevent collisions with the environment; in terms of computational efficiency, its geometric intuitiveness helps

reduce problem complexity; regarding solution success rate, constraining the search space enables more reliable identification of feasible solutions; and most critically, in terms of solution continuity, arm-angle-based parameterization can generate smooth, jerk-free joint trajectories, which is essential for high-safety on-orbit operations. However, as previously analyzed, these advantages hinge on one fundamental prerequisite—the method must be successfully applicable to the target manipulator's configuration. The 7-DOF manipulator with multiple offsets studied in this paper represents a typical scenario where traditional arm-angle methods fail. The complex spatial relationships among its shoulder, elbow, and wrist joint axes prevent the direct application of conventional shoulder point (S) and wrist point (W) definitions, thereby nullifying all the aforementioned advantages. Thus, resolving the "inapplicability" of the arm-angle method to this configuration is not only key to achieving inverse kinematics solutions but also a prerequisite for introducing core benefits such as obstacle avoidance, high efficiency, high success rate, and motion continuity to this class of manipulators. This study is dedicated to breaking through this bottleneck. We have successfully extended the arm-angle-based method to this manipulator configuration, filling a gap in this research area. This not only provides a novel and efficient solution for the inverse kinematics of the studied manipulator but, more importantly, lays a crucial theoretical and methodological foundation for its subsequent application in complex tasks requiring high-performance intelligent control.

Analytical inverse kinematics solutions are widely recognized for their computational efficiency. However, for the 7-DOF manipulator configuration studied in this paper, such closed-form solutions remain unavailable due to the inherent complexity of its kinematic structure.

For on-orbit servicing tasks, the accuracy, efficiency, and stability of manipulator IK are critical. While arm angle parameterization methods demonstrate geometric intuitiveness with unique advantages in configuration control optimization and obstacle avoidance, prior research has predominantly focused on Spherical-Revolute-Spherical (SRS) manipulators. In contrast, inverse kinematics for manipulators with concurrent shoulder-elbow-wrist offsets presents more complexity and technical challenges. Existing arm angle-based solutions for SSRMS manipulators are ineffective for the manipulator architecture under investigation in this study. Current research on similar architectures predominantly relies on purely numerical approaches, with limited exploration of arm angle-based methodologies.

This study first addresses the problem that traditional arm angle methods cannot define shoulder and wrist points due to the permanently non-coplanar and non-intersecting axes of joint 1 and joint 2, joint 6 and joint 7 in this configuration. By extending the rotation axes of joint 1 and joint 3, as well as the rotation axes of joint 6 and joint 7, a new definition method for shoulder and wrist points is pro-

posed. Different from previous concepts of approximate manipulator or virtual manipulator, this paper proposes the concept of a variable SRS equivalent manipulator and analyzes its characteristics and advantages. Based on this equivalent manipulator concept, the complex IK problem of the 7-DOF manipulator with simultaneous shoulder-elbow-wrist offsets is transformed into solving two nonlinear equations with two unknowns joint angle 2 (θ_2) and joint angle 6 (θ_6) of the real manipulator. Additionally, for solving these nonlinear equations, this paper proposes an initial value calculation method based on approximate manipulator and a hybrid solving method combining improved Levenberg-Marquard (LM) and improved PSO algorithms.

The remainder of the paper is arranged as follows. Section 2 establishes the kinematic model of the 7-DOF manipulator with multiple joint offsets. Section 3 proposes the equivalent manipulator concept and derives the joint angle solutions for the real manipulator based on arm angle parameterization, resulting in two nonlinear equations with θ_2 and θ_6 . Section 4 presents a hybrid nonlinear equation-solving method combining improved LM and improved PSO algorithms. The remaining joint angles of the real manipulator are determined by integrating the solved θ_2 and θ_6 with the equivalent manipulator concept. Section 5 validates the method's effectiveness and advantages through experiments and comparative analyses. Section 6 concludes the study and discusses future research directions.

This paper proposes a new method for solving the inverse kinematics of 7-DOF manipulators with multiple offsets. Its innovations are mainly reflected in the following three interrelated and progressively advanced aspects:

(1) Theoretical and conceptual innovation: a new concept of "Variable SRS-Equivalent Manipulator".

Addressing the fundamental bottleneck that traditional arm-angle methods cannot define shoulder/wrist points due to non-coplanar and non-intersecting joint axes, this paper proposes a new definition of shoulder and wrist points by extending the joint rotation axes. Unlike previous methods based on a fixed "approximate arm" or "virtual arm," the core of this work lies in the fact that the Modified Denavit-Hartenberg (MDH) parameters of the constructed SRS-equivalent manipulator are not fixed but vary dynamically with the desired end-effector pose. Each pose corresponds to a kinematically fully equivalent SRS configuration with unique MDH parameters. This conceptual breakthrough opens a new pathway for applying the mature arm-angle theory to the specific complex manipulator configuration studied herein.

(2) Solver framework innovation: a dimensionality-reduction solving framework.

Based on the above concept of a variable equivalent manipulator and the arm-angle parameterization method, this work transforms the complex 7-DOF inverse kinematics problem under joint motion constraints into solving a low-dimensional system of nonlinear equations with only

two unknowns (θ_2, θ_4) within their feasible domain. This framework fundamentally resolves the "incompatibility" between traditional arm-angle methods and the manipulator configuration studied here, significantly reducing computational dimensionality and cost, and laying the groundwork for efficient, high-precision, and high-safety solutions.

(3) Algorithmic innovation: an efficient hybrid improved solving algorithm.

To solve the aforementioned nonlinear system, this paper proposes a hybrid strategy that deeply integrates an improved LM algorithm and an improved PSO algorithm. The originality of this algorithm goes beyond a simple combination of methods. By introducing various dynamic parameter adjustment strategies and innovatively proposing an initial value calculation method based on an SRS approximate model, it effectively combines the advantages of optimization algorithms and intelligent algorithms, thereby significantly improving the solution success rate and computational speed. This provides a robust numerical tool for handling such complex inverse kinematics problems.

By leveraging the above innovations, this study successfully introduces the arm angle as a redundant parameter into the solution process, thereby filling the research gap in applying the arm-angle method to the studied manipulator configuration. This breakthrough establishes a method that simultaneously meets the stringent on-orbit requirements for high solving stability, high computational efficiency, and high solution accuracy—a set of capabilities beyond the reach of existing state-of-the-art algorithms. The subsequent chapters will delve into the details of this methodology.

2. Kinematic model of a 7-DOF robotic manipulator

The multi-offset 7-DOF robotic manipulator is a revolute-joint-only robotic manipulator featuring shoulder offset a_1 , elbow offset a_3 , and wrist offset a_5 . This configuration is exemplified by practical implementations such as the European Space Agency's "MIRROR" project (Deremetz et al., 2023) and ABB's YuMi collaborative robot system (Asgari et al., 2025). As illustrated in Fig. 1, its initial configuration exhibits collinear alignment of joint axes 1, 3, 5, and 7. The MDH coordinate system is established as shown in Fig. 1. In this paper, the $\sum_i$ ($i = 0, 1, \dots, 7$) coordinate system is defined with x_i ($i = 0, 1, \dots, 7$) and z_i ($i = 0, 1, \dots, 7$) representing its x-axis and z-axis. Table 1 details the MDH parameters of the multi-offset 7-DOF manipulator, where d_i ($i = 1, 2, \dots, 7$) and θ_i ($i = 1, 2, \dots, 7$) specify translation and rotation about the z_i ($i = 1, 2, \dots, 7$), while a_{i-1} ($i = 1, 2, \dots, 7$) and α_{i-1} ($i = 1, 2, \dots, 7$) define translation and rotation about the x_{i-1} ($i = 1, 2, \dots, 7$).

The transformation matrix from coordinate system $\sum_i$ to $\sum_{i-1}$ can be calculated using Eq. (1).

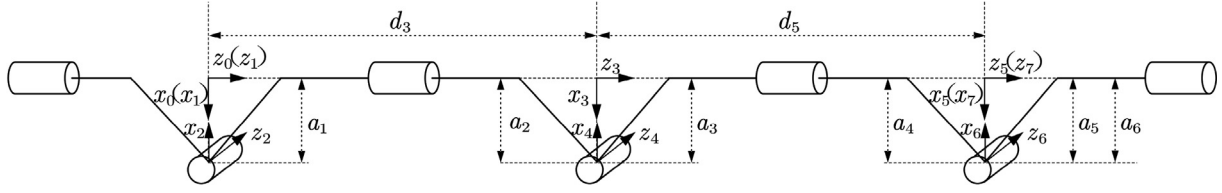


Fig. 1. The kinematic model.

Table 1
MDH parameters of the multi-offset 7-DOF manipulator.

Joint	$a_{i-1}/(m)$	$\alpha_{i-1}/(^{\circ})$	$d_i/(m)$	$\theta_i/(^{\circ})$	Range/ $(^{\circ})$
1	0	0	0	$\theta_1 - 180$	$-170 \sim 170$
2	0.066	-90	0	$\theta_2 - 180$	$-42.5 \sim 170$
3	0.066	-90	0.53	$\theta_3 - 180$	$-170 \sim 170$
4	0.066	-90	0	$\theta_4 - 180$	$-42.5 \sim 170$
5	0.066	-90	0.53	$\theta_5 - 180$	$-170 \sim 170$
6	0.066	-90	0	$\theta_6 - 180$	$-42.5 \sim 170$
7	0.066	-90	0	$\theta_7 - 180$	$-170 \sim 170$

$${}^{i-1}T_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i & 0 & a_{i-1} \\ \sin \theta_i \cos \alpha_{i-1} & \cos \theta_i \cos \alpha_{i-1} & -\sin \alpha_{i-1} & -\sin \alpha_{i-1} d_i \\ \sin \theta_i \sin \alpha_{i-1} & \cos \theta_i \sin \alpha_{i-1} & \cos \alpha_{i-1} & \cos \alpha_{i-1} d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

In this paper $c_i = \cos \theta_i$, $s_i = \sin \theta_i$.

$${}^0T_7 = {}^0T_1 {}^1T_2 {}^2T_3 {}^3T_4 {}^4T_5 {}^5T_6 {}^6T_7 = \begin{bmatrix} {}^0R_7 & {}^0P_7 \\ 0 & 1 \end{bmatrix} \quad (2)$$

0R_7 and 0P_7 represent the rotation matrix and position vector of the coordinate system $\sum_7$ relative to the base coordinate system $\sum_0$, and 0T_7 denotes the desired end-effector pose.

3. An inverse kinematics solving method for the multi-offset 7-DOF manipulator based on variable SRS equivalent manipulator and arm angle

3.1. The concept of variable SRS equivalent manipulator

As shown in the Fig. 2, for joints 1, 2, and 3, when θ_2 is not equal to $\pm 180^\circ$, the extended line of the rotational axis of joint 3 (z_3) always intersects the rotational axis of joint 1 (z_1) regardless of the values of the seven joint angles (when $\theta_2=0^\circ$, z_1 coincides with z_3), with the intersection point denoted as S . For the rotation of joint 2, the motion of S_0' corresponds to circular movement with radius $S_0'O_2 = S_0O_2$ around center O_2 , where the magnitude of S_0O_2 remains constant and equals the MDH parameters a_1 and a_2 . Thus, in triangle SS_0O_2 , the relationship in Eq. (3) holds.

$$S'_0S = S_0S = S_0O_2 \tan \frac{\theta_2}{2} \quad (3)$$

By the same reasoning, for joints 5, 6, and 7. When θ_6 is not equal to $\pm 180^\circ$, the extended line of the rotational axis of joint 7 (z_7) always intersects the rotational axis of joint 5 (z_5) (when $\theta_6=0^\circ$, z_5 coincides with z_7), defining intersection W . The motion of W_0' during joint 6 rotation follows a circular trajectory with radius $W_0'O_6 = W_0O_6 = a_6 = a_7$ around O_6 . Consequently, the relationship in Eq. (4) persists during motion.

$$W'_0W = W_0W = W_0O_6 \tan \frac{\theta_6}{2} \quad (4)$$

The non-coplanar and non-intersecting axes of joints 1–2 and joints 6–7 in the real manipulator render traditional shoulder/wrist point definitions based on arm angle methods inapplicable. In the paper, the shoulder point S and wrist point W of the variable SRS equivalent manipulator are selected as the true shoulder and wrist points. As illustrated, this configuration forms a variable SRS equivalent manipulator sharing identical end-effector poses. Crucially, the MDH parameters of the variable SRS equivalent manipulator vary with different target poses and arm angles, constituting one of the core innovations of the proposed methodology. Existing arm-angle-parameterized methods aim to identify a unique approximated SRS manipulator. These methods solve for candidate joint angles through the approximated SRS configuration, establish mathematical correspondences between the real manipulator and the approximated SRS manipulator via joint-angle parameterization features, and refine solutions through error correction. However, such approaches are incompatible with the manipulator studied here due to two inherent limitations:

- No unique approximated SRS manipulator can be identified;
- Mathematical relationships between the joint angles of the real manipulator and the approximated SRS manipulator are intractable.

The critical aspect of the proposed method lies in establishing a SRS equivalent manipulator rather than an approximated one. Equivalence is defined as identical end-effector poses between the real manipulator and the SRS equivalent manipulator when all seven joint angles are identical.

As shown in the Fig. 3, to ensure identical end-effector poses between the SRS equivalent manipulator and the real multi-offset manipulator under identical joint angles, the

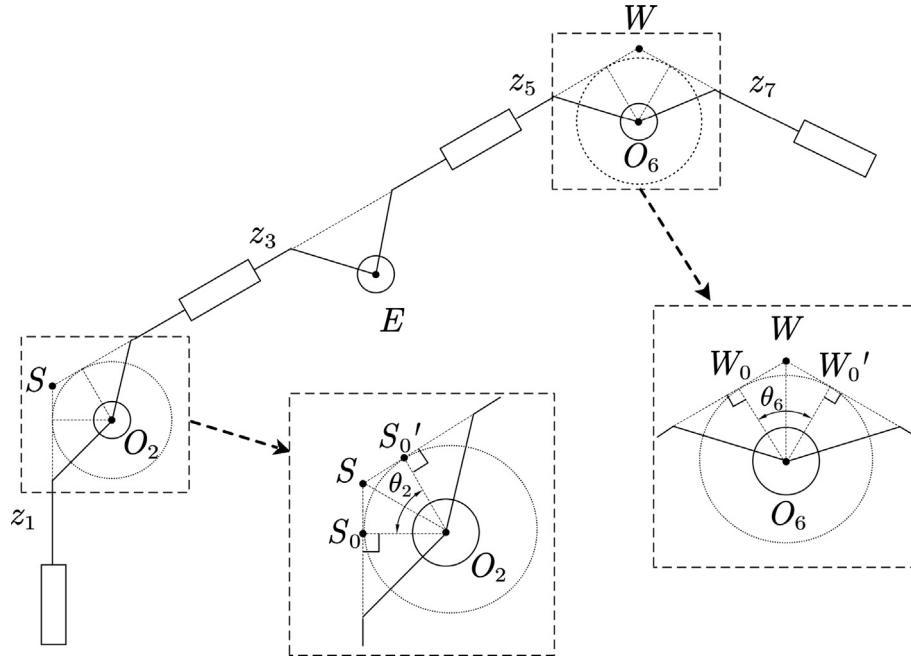


Fig. 2. Definition of shoulder and wrist points of the multi-offset 7-DOF manipulator and schematic of variable equivalent manipulator.

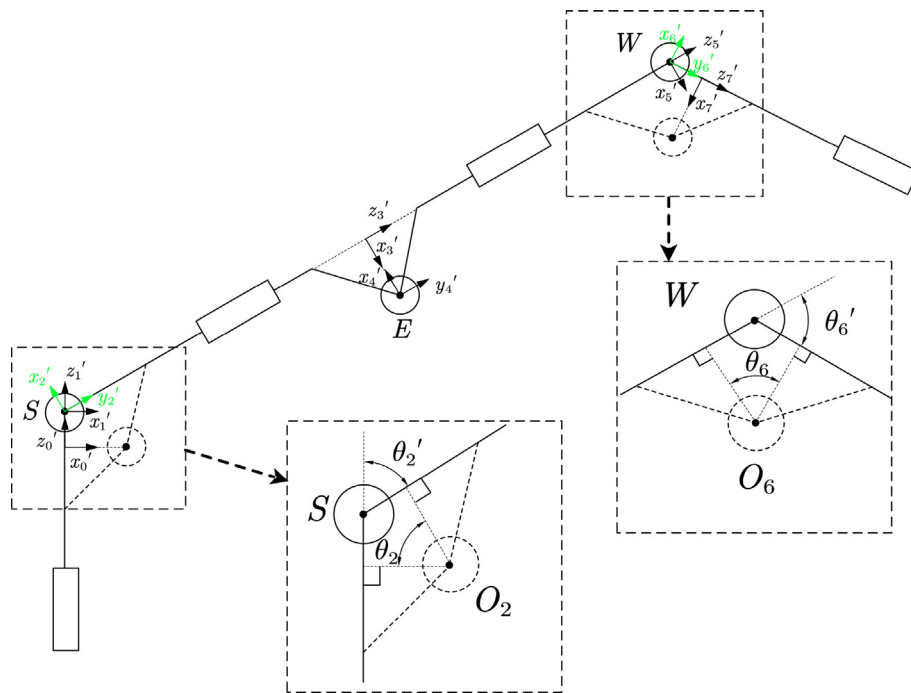


Fig. 3. Coordinate system diagram of variable equivalent manipulator. The dashed circle in the figure represents the joint position of the real manipulator, while the solid circle represents that of the equivalent manipulator.

coordinate systems $\sum_0$ and $\sum_7$ of the SRS equivalent manipulator must be constructed at the same positions as those of the actual multi-offset manipulator. Based on geometric relationships, we have:

$$\theta_2 = \theta'_2 \quad (5)$$

$$\theta_6 = \theta'_6 \quad (6)$$

θ'_2 as the joint 2 angle of the SRS equivalent manipulator, θ'_6 as the joint 6 angle of the SRS equivalent manipulator, $\sum'_i (i = 0, 1, \dots, 7)$ as the MDH coordinate system of the SRS equivalent manipulator, $x'_i (i = 0, 1, \dots, 7), y'_i (i = 0, 1, \dots, 7)$ and $z'_i (i = 0, 1, \dots, 7)$ as the x-axis, y-axis, and z-axis of $\sum'_i (i = 0, 1, \dots, 7)$, respectively. The SRS equivalent manipulator is no longer unique; instead, a dis-

Table 2
MDH parameters of the variable SRS equivalent manipulator.

Joint	$a_{i-1}/(m)$	$\alpha_{i-1}/(^{\circ})$	$d_i/(m)$	$\theta_i/(^{\circ})$	Range/ $(^{\circ})$
1	0	0	$0.066 \tan(\theta_2/2)$	$\theta_1 - 180$	$-170 \sim 170$
2	0	-90	0	$\theta_2 - 180$	$-42.5 \sim 170$
3	0	-90	$0.53 + 0.066 \tan(\theta_2/2)$	$\theta_3 - 180$	$-170 \sim 170$
4	0.066	-90	0	$\theta_4 - 180$	$-42.5 \sim 170$
5	0.066	-90	$0.53 + 0.066 \tan(\theta_6/2)$	$\theta_5 - 180$	$-170 \sim 170$
6	0	-90	0	$\theta_6 - 180$	$-42.5 \sim 170$
7	0	-90	$0.066 \tan(\theta_6/2)$	$\theta_7 - 180$	$-170 \sim 170$

tinct SRS equivalent manipulator corresponds to each target pose to be solved. Specifically, the MDH parameters of the SRS equivalent manipulator are not arbitrarily different but can be expressed through θ_2 and θ_6 . As shown in Table 2, the adjustment essentially displaces the link of joint 1 along the positive direction of $\hat{z}_1$ by $S_0 O_2 \tan(\theta_2/2)$, the link of joint 3 along the negative direction of $\hat{z}_3$ by $S_0 O_2 \tan(\theta_2/2)$, the link of joint 5 along the positive direction of $\hat{z}_5$ by $W_0 O_6 \tan(\theta_6/2)$, and the link of joint 7 along the negative direction of $\hat{z}_7$ by $W_0 O_6 \tan(\theta_6/2)$. It is important to note that θ_2 and θ_6 are not predefined but are incorporated as unknown variables to be solved within the MDH parameters of the dynamic SRS equivalent manipulator, as detailed in Table 2.

As shown in Fig. 4, the equivalent schematic diagrams correspond to θ_2 within the intervals of $[-90^{\circ}, 0^{\circ})$, 0° , $(0^{\circ}, 90^{\circ})$, $[90^{\circ}, 180^{\circ})$, 180° and $(180^{\circ}, 270^{\circ})$, respectively.

As shown in Fig. 4(f), the scenario for θ_2 angles within $180^{\circ} \sim 270^{\circ}$ presents a unique case. Although this range exceeds the joint limits of θ_2 , the proposed method remains capable of deriving valid solutions. In this configuration, the equivalent manipulator adopts a kinematically valid but physically impractical structure. Specifically, joint 1 is positioned between points *A* and *B*, with a rigid link extending from point *B* through joint 1 along its axis to point *S* (the location of joint 2). Another rigid link connects point *S* to point *C*, where the axis of joint 3 coincides with the link between *S* and *C*. Joint 3 spans from point *C* to *D*, followed by joint 4 at point *O*₄. While such a configuration would be mechanically infeasible in real-world manipulators, it serves as an effective mathematical abstraction within the equivalent manipulator framework for inverse kinematics resolution. This demonstrates the methodology's applicability to manipulators with identical kinematic structures but differing joint limit constraints. Similarly, the equivalence principle governing θ_6 is identical to that of θ_2 .

As shown in Fig. 4(b), when θ_2 is at 0° , the axes z'_1 and z'_3 become coincident. The selection of point *S* under this configuration is illustrated, while the selection of point *W* follows an analogous principle when θ_6 is at 0° . For cases where θ_2 or θ_6 reaches $\pm 180^{\circ}$, both exceed standard joint limits, as show in Fig. 4(e). In these extreme configurations, points *S* or *W* theoretically lie at infinity, causing specific MDH parameters of the equivalent manipulator to

approach infinite values. If practical applications require these angles, θ_2 or θ_6 can be treated as predetermined inputs, enabling efficient computation of remaining joint angles through parametric methods. However, such scenarios (θ_2 or θ_6 at $\pm 180^{\circ}$) are statistically negligible in general operations. Most manipulators of this architecture inherently restrict θ_2 and θ_6 within non-singular ranges through mechanical design. Consequently, these singular configurations exhibit minimal impact on inverse kinematics success rates, as they reside beyond the effective workspace of practical implementations.

3.2. Determination of θ_4

For the shoulder-elbow-wrist (*S-E-W*) points, when $\theta_4 = 0^{\circ}$, it is defined as the initial state S_0 - E_0 - W_0 . As shown in Fig. 5, the length of $S_0 E_0$ can be represented as:¹

$$S_0 E_0 = \sqrt{a_2^2 + \left[d_3 + a_1 \tan \frac{\theta_2}{2} \right]^2} \quad (7)$$

$$E_0 W_0 = \sqrt{a_3^2 + \left[d_5 + a_4 \tan \frac{\theta_6}{2} \right]^2} \quad (8)$$

$$S_0 W_0 = d_3 + a_1 \tan \frac{\theta_2}{2} + d_5 + a_4 \tan \frac{\theta_6}{2} \quad (9)$$

In the initial state S_0 - E_0 - W_0 :

$$\angle S_0 E_0 W_0 = \arccos \frac{S_0 E_0^2 + E_0 W_0^2 - S_0 W_0^2}{2 \cdot S_0 E_0 \cdot E_0 W_0} \quad (10)$$

Next, θ_4 is calculated. When θ_4 is not 0° , in the actual *S-E-W*, compared to the initial state S_0 - E_0 - W_0 :

$$\begin{cases} SE = S_0 E_0 \\ EW = E_0 W_0 \end{cases} \quad (11)$$

$$SW = ||\mathbf{SW}|| \quad (12)$$

$$\mathbf{SW} = \mathbf{OW} - \mathbf{OS} \quad (13)$$

The coordinates of point *S* are $(0, 0, -a_1 \tan(\theta_2/2))$ and the direction vector $\hat{z}_7$ is $({}^0 T_7(1, 3), {}^0 T_7(2, 3), {}^0 T_7(3, 3))$ where ${}^0 T_7(i, j)$ represents the element in the *i*-th row and *j*-th column of ${}^0 T_7$. The coordinates of point *W* are:

¹ In the following paper, if the coordinate system of a vector or a point is $\sum_0'$, then the superscript is omitted for simplicity.

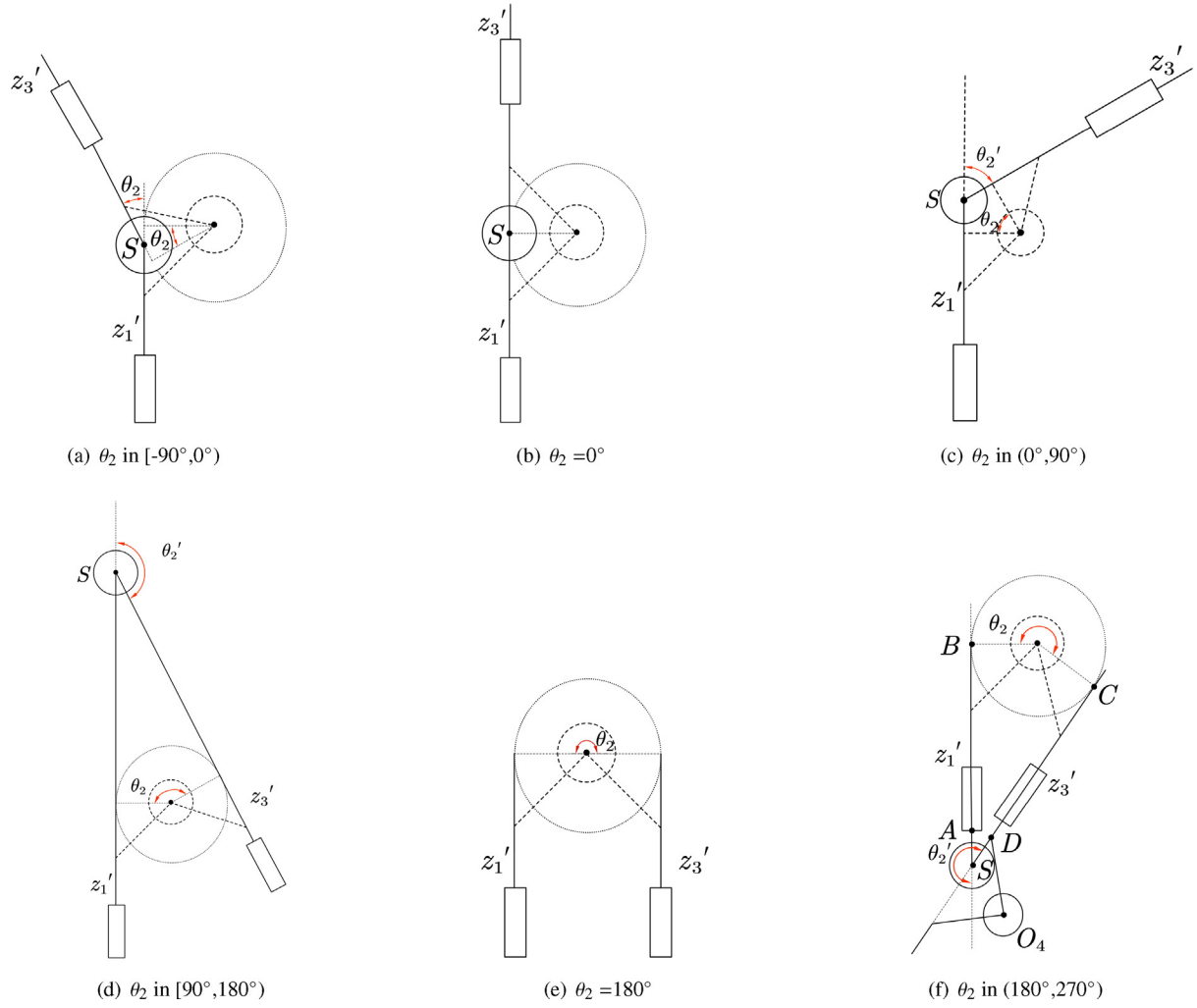


Fig. 4. Equivalent diagram of joint 2. The small dashed circle in the figure indicates the position of joint 2 of the real manipulator, while the solid circle shows that of joint 2 of the equivalent manipulator.

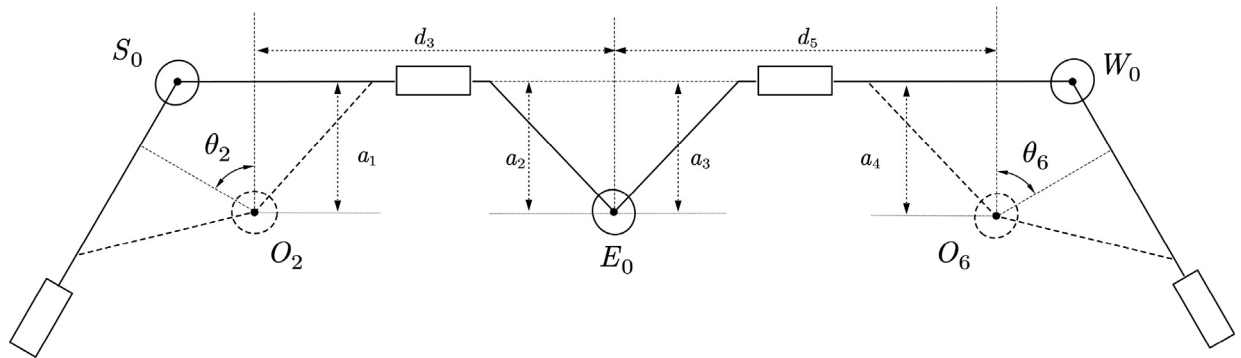


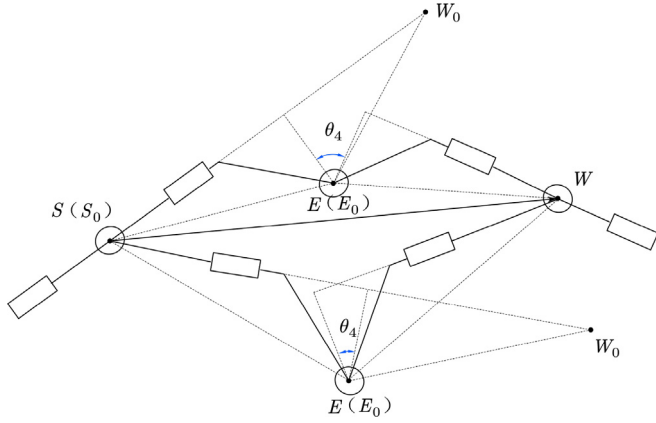
Fig. 5. The initial state S_0 - E_0 - W_0 of the variable equivalent manipulator.

$$W = \begin{pmatrix} {}^0T_7(1, 4) - \left(d_7 + a_6 \tan \frac{\theta_6}{2}\right) {}^0T_7(1, 3), {}^0T_7(2, 4) \\ - \left(d_7 + a_6 \tan \frac{\theta_6}{2}\right) {}^0T_7(2, 3), {}^0T_7(3, 4) - \left(d_7 + a_6 \tan \frac{\theta_6}{2}\right) {}^0T_7(3, 3) \end{pmatrix} \quad (14)$$

$$SW = \begin{pmatrix} {}^0T_7(1, 4) - \left(d_7 + a_6 \tan \frac{\theta_6}{2}\right) {}^0T_7(1, 3) \\ {}^0T_7(2, 4) - \left(d_7 + a_6 \tan \frac{\theta_6}{2}\right) {}^0T_7(2, 3) \\ {}^0T_7(3, 4) - \left(d_7 + a_6 \tan \frac{\theta_6}{2}\right) {}^0T_7(3, 3) + S_0 O_2 \cdot \tan \frac{\theta_2}{2} \end{pmatrix} \quad (15)$$

Therefore, the coordinate of the vector SW relative to $\sum_0'$ can be expressed as:

As a result, SW can be expressed in terms of θ_2 and θ_6 . In the actual S - E - W :

Fig. 6. Two possible solutions of θ_4 for each $\mathbf{SW}$.

$$\angle SEW = \arccos \frac{SE^2 + EW^2 - SW^2}{2 \cdot SE \cdot EW} \quad (16)$$

As shown in Fig. 6, for each $\mathbf{SW}$, θ_4 has two possible solutions:

$$\theta_4 = \angle SEW - \angle S_0 E_0 W_0 \quad (17)$$

$$\theta_4 = 2\pi - \angle SEW - \angle S_0 E_0 W_0 \quad (18)$$

3.3. Arm angle based determination of initial pose ${}^0R_3^{\psi=0}$

As shown in Fig. 7, let $\hat{v} = (0, 0, 1)$ denote the rotation axis vector of joint 1. The reference plane is defined by vectors $\hat{v}$ and $\mathbf{SW}$, while the arm plane is formed by vectors $\mathbf{SW}$ and $\mathbf{SE}$. The arm angle ψ is the rotation angle required to align the reference plane with the arm plane by rotating the reference plane around $\mathbf{SW}$ according to the right-hand rule. Here, ${}^0R_3^{\psi=0}$ represents the orientation matrix of coordinate 3 when $\psi = 0^\circ$.

As illustrated in Fig. 8 and governed by the MDH convention, the following equation holds universally regardless of the seven joint angles of the manipulator:

$$\mathbf{SE} = a_2 \mathbf{x}'_3 + \left(d_3 + a_1 \tan \frac{\theta_2}{2} \right) \mathbf{z}'_3 \quad (19)$$

$$\mathbf{EW} = a_3 \mathbf{x}'_4 + \left(d_5 + a_5 \tan \frac{\theta_6}{2} \right) \mathbf{y}'_4 \quad (20)$$

$$\mathbf{EW}^0 = a_3 {}^0\mathbf{x}_4^0 + \left(d_5 + a_5 \tan \frac{\theta_6}{2} \right) {}^0\mathbf{y}_4^0 \quad (22)$$

${}^0R_4^{\psi=0} = [{}^0\mathbf{x}_4^0 \ {}^0\mathbf{y}_4^0 \ {}^0\mathbf{z}_4^0]$ denotes the orientation matrix of $\sum_4'$ relative to $\sum_0'$ when $\psi = 0^\circ$, and ${}^0R_3^{\psi=0} = [{}^0\mathbf{x}_3^0 \ {}^0\mathbf{y}_3^0 \ {}^0\mathbf{z}_3^0]$ represents the orientation matrix of $\sum_3'$ relative to $\sum_0'$ when $\psi = 0^\circ$. The vector $\mathbf{EW}^0$ corresponds to $\mathbf{EW}$ if $\psi = 0^\circ$, while $\mathbf{SE}^0$ defines $\mathbf{SE}$ under the same arm angle condition. Based on the orientation transformation relationship:

$${}^3R_4 = \begin{bmatrix} -c_4 & s_4 & 0 \\ 0 & 0 & 1 \\ s_4 & c_4 & 0 \end{bmatrix} = [{}^3\mathbf{x}_4 \ {}^3\mathbf{y}_4 \ {}^3\mathbf{z}_4] \quad (23)$$

The following equalities hold:

$${}^0\mathbf{x}_4^0 = {}^0R_3^{\psi=0} {}^3\mathbf{x}_4 = [{}^0\mathbf{x}_3^0 \ {}^0\mathbf{y}_3^0 \ {}^0\mathbf{z}_3^0] {}^3\mathbf{x}_4 = -c_4 {}^0\mathbf{x}_3^0 + s_4 {}^0\mathbf{z}_3^0 \quad (24)$$

$${}^0\mathbf{y}_4^0 = {}^0R_3^{\psi=0} {}^3\mathbf{y}_4 = [{}^0\mathbf{x}_3^0 \ {}^0\mathbf{y}_3^0 \ {}^0\mathbf{z}_3^0] {}^3\mathbf{y}_4 = s_4 {}^0\mathbf{x}_3^0 + c_4 {}^0\mathbf{z}_3^0 \quad (25)$$

Next, calculate $\mathbf{SE}^0$. When $\psi = 0^\circ$, the vector $\mathbf{SE}^0$ can be obtained by rotating $\mathbf{SW}$ around the vector $\mathbf{l}$ with the angle α . Therefore, $\mathbf{SE}^0 = (||\mathbf{SE}^0|| / ||\mathbf{SW}||) \cdot R(\mathbf{l}, \alpha) \cdot \mathbf{SW}$, where $\mathbf{l} = (1 / ||\mathbf{SW}||) \cdot \mathbf{v} \times \mathbf{SW}$, $\alpha = \arccos((SW^2 + SE^2 - EW^2) / (2 \cdot SW \cdot SE))$, $R(\mathbf{l}, \alpha) = I_3 + [u_l \times] \sin \alpha + [u_l \times]^2 (1 - \cos \alpha)$.

Where $[u_l \times]$ is the skew-symmetric matrix of vector $\mathbf{l}$. Assume that $\mathbf{l} = [l_x, l_y, l_z]$, then:

$$[u_l \times] = \begin{bmatrix} 0 & -l_z & l_y \\ l_z & 0 & -l_x \\ -l_y & l_x & 0 \end{bmatrix} \quad (26)$$

Thereby, vector $\mathbf{EW}^0$ can be expressed as $\mathbf{EW}^0 = \mathbf{SW} - \mathbf{SE}^0$.

Given that $\mathbf{SE}^0$ and $\mathbf{EW}^0$ is known, combine the Eqs. (21), (22), (24) and (25), we can derive ${}^0\mathbf{x}_3^0$ and ${}^0\mathbf{z}_3^0$:

$${}^0\mathbf{x}_3^0 = \frac{\mathbf{SE}^0 - (d_3 + a_1 \tan \frac{\theta_2}{2}) {}^0\mathbf{z}_3^0}{a_3} \quad (27)$$

$${}^0\mathbf{z}_3^0 = \frac{(-a_4 c_4 + (d_5 + a_5 \tan \frac{\theta_6}{2}) s_4 + a_3) \mathbf{SE}^0 - a_3 \mathbf{SW}}{-a_3 a_4 s_4 - a_3 (d_5 + a_5 \tan \frac{\theta_6}{2}) c_4 - a_4 (d_3 + a_1 \tan \frac{\theta_2}{2}) c_4 + (d_3 + a_1 \tan \frac{\theta_2}{2}) (d_5 + a_5 \tan \frac{\theta_6}{2}) s_4} \quad (28)$$

When the arm angle $\psi = 0^\circ$, the relationships stated above remain valid. Therefore, the following holds:

$$\mathbf{SE}^0 = a_2 {}^0\mathbf{x}_3^0 + \left(d_3 + a_1 \tan \frac{\theta_2}{2} \right) {}^0\mathbf{z}_3^0 \quad (21)$$

According to the right-hand rule, ${}^0\mathbf{y}_3^0 = {}^0\mathbf{z}_3^0 \times {}^0\mathbf{x}_3^0$. ${}^0R_3^{\psi=0}$ is finally expressed as ${}^0R_3^{\psi=0} = [{}^0\mathbf{x}_3^0 \ {}^0\mathbf{y}_3^0 \ {}^0\mathbf{z}_3^0]$, which is a function of $\theta_2, \theta_4, \theta_6$.

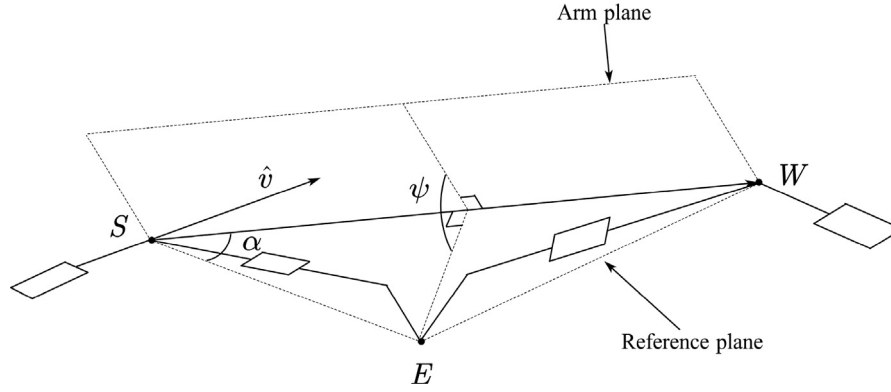


Fig. 7. Definition of the arm angle.

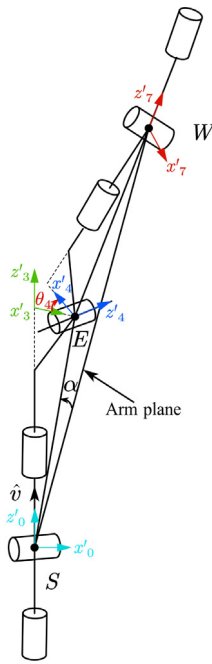


Fig. 8. Equivalent manipulator with coordinate.

3.4. Determination of the shoulder angle θ_2

The real rotation matrix 0R_3 can be derived by revolving the ${}^0R_3^{\psi=0}$ around the vector $\mathbf{SW}$ with the arm angle ψ . According to the Rodrigues formula, ${}^0R_\psi = I_3 + [{}^0u_w \times] \sin \psi + [{}^0u_w \times]^2 (1 - \cos \psi)$. Consequently, the rotation matrix of $\sum_3'$ relative to $\sum_0'$ can be calculated as:

$${}^0R_3 = {}^0R_\psi \cdot {}^0R_3^{\psi=0} = \begin{bmatrix} a_{s11} & a_{s12} & a_{s13} \\ a_{s21} & a_{s22} & a_{s23} \\ a_{s31} & a_{s32} & a_{s33} \end{bmatrix} \quad (29)$$

where $a_{sij}(i, j = 1, 2, 3)$ represents the element of 0R_3 on the i -th row, j -th column. As mentioned above, $a_{sij}(i, j = 1, 2, 3)$ is essentially a non-linear expression of $\theta_2, \theta_4, \theta_6$, where $a_{s33} = f(\theta_2, \theta_4, \theta_6)$.

From another perspective, according to the homogeneous transformation formula in the MDH coordinate:

$${}^0R_3 = {}^0R_1 {}^1R_2 {}^2R_3 = \begin{bmatrix} c_1 c_2 c_3 - s_1 s_3 & -c_1 c_2 s_3 - s_1 c_3 & c_1 s_2 \\ s_1 c_2 c_3 + c_1 s_3 & -s_1 c_2 s_3 + c_1 c_3 & s_1 s_2 \\ -s_2 c_3 & s_2 s_3 & c_2 \end{bmatrix} \quad (30)$$

where the element $c_2 = \cos \theta_2 = a_{s33} = f(\theta_2, \theta_4, \theta_6)$. Hence:

$$\theta_2 = \pm \arccos a_{s33} = \pm \arccos f(\theta_2, \theta_4, \theta_6) \quad (31)$$

3.5. Determination of the wrist angle θ_6

The expression of 4R_7 is derived firstly. Based on homogeneous transformation, ${}^7R_4 = {}^0R_7^T \cdot {}^0R_\psi \cdot {}^0R_3^{\psi=0} \cdot {}^3R_4$, ${}^4R_7 = {}^7R_4^T$. Denote each element of the above matrix ${}^4R_7 = \begin{bmatrix} a_{t11} & a_{t12} & a_{t13} \\ a_{t21} & a_{t22} & a_{t23} \\ a_{t31} & a_{t32} & a_{t33} \end{bmatrix}$, where $a_{tij}(i, j = 1, 2, 3)$ represents to the element on the i -th row, j -th column. Similarly, $a_{tij}(i, j = 1, 2, 3)$ is a non-linear expression of $\theta_2, \theta_4, \theta_6$, where $a_{t23} = g(\theta_2, \theta_4, \theta_6)$.

In the same way, we can calculate 4R_7 by the MDH coordinate:

$${}^4R_7 = {}^4R_5 {}^5R_6 {}^6R_7 = \begin{bmatrix} -c_5 c_6 c_7 + s_5 s_7 & c_5 c_6 s_7 + s_5 c_7 & -c_5 s_6 \\ -s_6 c_7 & s_6 s_7 & c_6 \\ s_5 c_6 c_7 + c_5 s_7 & -s_5 c_6 s_7 + c_5 c_7 & s_5 s_6 \end{bmatrix} \quad (32)$$

where $c_6 = \cos \theta_6 = a_{t23} = g(\theta_2, \theta_4, \theta_6)$. Therefore:

$$\theta_6 = \pm \arccos a_{t23} = \pm \arccos g(\theta_2, \theta_4, \theta_6). \quad (33)$$

Since there are all two solutions for $\theta_2, \theta_4, \theta_6$ as shown in Eqs. (17), (18), (31) and (33), we can get eight different simultaneous equations, each including three non-linear equations of $\theta_2, \theta_4, \theta_6$. Based on the above analysis, the IK problem of the multi-offset 7-DOF manipulator can be formulated as the solution of the non-linear simultaneous equations with the parameters of $\theta_2, \theta_4, \theta_6$. Notably, different from previous research according to arm angle,

the method suggested in the paper can obtain accurate solutions of $\theta_2, \theta_4, \theta_6$ free from the need of further modification.

3.6. Determination of the rest angles $\theta_1, \theta_3, \theta_5, \theta_7$

By substituting $\theta_2, \theta_4, \theta_6$ into Eqs. (30) and (32), the rotation matrix 0R_3 and 4R_7 can be determined, and the rest angles can be obtained by solving Euler angles. Specifically, for θ_1 and θ_3 , comparing the two expressions of 0R_3 yields the following: $c_1s_2 = a_{s13}, s_1s_2 = a_{s23}, -s_2c_3 = a_{s31}$, and $s_2s_3 = a_{s32}$. Consequently, θ_1 and θ_3 can be obtained as:

$$\theta_1 = \arctan 2(a_{s23}s_2, a_{s13}s_2) \quad (34)$$

$$\theta_3 = \arctan 2(a_{s32}s_2, -a_{s31}s_2) \quad (35)$$

Similarly, for θ_5 and θ_7 , comparing the two expressions of 4R_7 yields $-c_5s_6 = a_{t13}, s_5s_6 = a_{t33}, -s_6c_7 = a_{t21}, s_6s_7 = a_{t22}$. Thus, the following equation holds:

$$\theta_5 = \arctan 2(a_{t33}s_6, -a_{t13}s_6) \quad (36)$$

$$\theta_7 = \arctan 2(a_{t22}s_6, -a_{t21}s_6) \quad (37)$$

As shown in Eqs. (17) and (18), θ_4 can be expressed in terms of θ_2 and θ_6 . Therefore $\theta_1, \theta_3, \theta_5$, and θ_7 can be expressed in terms of θ_2 and θ_6 .

4. An efficient hybrid algorithm integrating the improved LM and the improved PSO with a novel initialization strategy

Firstly, the substitution of Eqs. (17) and (18) into Eqs. (31) and (33) eliminates θ_4 , yielding two coupled nonlinear equations:

$$F(\theta_2, \theta_6) = 0 \quad (38)$$

$$G(\theta_2, \theta_6) = 0 \quad (39)$$

This reduces the original problem of solving for three joint angles $(\theta_2, \theta_4, \theta_6)$ to a problem of solving a system of two nonlinear equations in two unknowns (θ_2, θ_6) .

Currently, main methods for solving nonlinear equation groups are traditional optimization algorithms, intelligent algorithms, and hybrid algorithms (Sharma and Kumar, 2022). Traditional algorithms have advantages of fast computing speed and mature theory, but they are sensitive to initial values and prone to falling into local optima. Intelligent algorithms have strong global search ability and are suitable for multi-peak problems, but they have large computational loads and slow convergence (Li et al., 2021b). Hybrid algorithms combine the advantages of global search and local fast convergence, showing good stability in solving complex nonlinear problems (Rajwar et al., 2023). To overcome the limitations of single algorithms, this paper proposes a hybrid algorithm based on an improved Levenberg Marquardt algorithm and an improved PSO algorithm.

4.1. The improved LM algorithm

The improved LM algorithm (Chen and Ma, 2023) is a method between gradient descent and Newton's method. It combines the stability of gradient descent and the fast convergence of Newton's method, and optimizes the parameter selection strategy of LM. It is suitable for nonlinear least squares problems.

To solve the system of equations, we define the function:

$$H(x) = \begin{pmatrix} F(\theta_2, \theta_6) \\ G(\theta_2, \theta_6) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (40)$$

where $x = \begin{pmatrix} \theta_2 \\ \theta_6 \end{pmatrix}$. At the current iteration point $x_k = \begin{pmatrix} \theta_{2(k)} \\ \theta_{6(k)} \end{pmatrix}$, the function value is: $H(x_k) = \begin{pmatrix} F(\theta_{2(k)}, \theta_{6(k)}) \\ G(\theta_{2(k)}, \theta_{6(k)}) \end{pmatrix}$.

The Jacobian matrix J_k is the gradient matrix of $H(x_k)$ at x_k :

$$J_k = \begin{pmatrix} \frac{\partial F}{\partial \theta_{2(k)}} & \frac{\partial F}{\partial \theta_{6(k)}} \\ \frac{\partial G}{\partial \theta_{2(k)}} & \frac{\partial G}{\partial \theta_{6(k)}} \end{pmatrix} \quad (41)$$

Using the improved LM method, the LM parameter is:

$$\lambda_k = \gamma \|H_k\|^\delta + (1 - \gamma) \|J_k^T H_k\|^\delta \quad (42)$$

where:

$$\|H_k\| = \sqrt{F(\theta_{2(k)}, \theta_{6(k)})^2 + G(\theta_{2(k)}, \theta_{6(k)})^2} \quad (43)$$

$$\|J_k^T H_k\| = \sqrt{\left(\frac{\partial F}{\partial \theta_{2(k)}} F + \frac{\partial G}{\partial \theta_{2(k)}} G \right)^2 + \left(\frac{\partial F}{\partial \theta_{6(k)}} F + \frac{\partial G}{\partial \theta_{6(k)}} G \right)^2} \quad (44)$$

The parameter $\gamma (\gamma \in [0, 1])$ balances $\|H_k\|^\delta$ and $\|J_k^T H_k\|^\delta$. The parameter $\delta (\delta \in [1, 2])$ adjusts the sensitivity of the parameters. Iteration stops when $\|J_k^T H_k\|^\delta \leq 10^{-6}$.

In each iteration, compute the search direction:

$$d_k = -(J_k^T J_k + \lambda_k I)^{-1} J_k^T H_k \quad (45)$$

Then update the iteration point:

$$x_{k+1} = x_k + d_k \quad (46)$$

To ensure solutions are within joint limits, constraints are added. As previously derived, $\theta_1, \theta_3, \theta_5, \theta_7$ can be expressed in terms of θ_2, θ_6 . Based on joint limits and Eqs. (34)–(37), there are 7 inequality constraints:

$$-170 \leq L_i(\theta_{2(k+1)}, \theta_{6(k+1)}) \leq 170 (i = 1, 3, 5, 7) \quad (47)$$

$$-42.5 \leq L_i(\theta_{2(k+1)}, \theta_{6(k+1)}) \leq 170 (i = 2, 4, 6) \quad (48)$$

where:

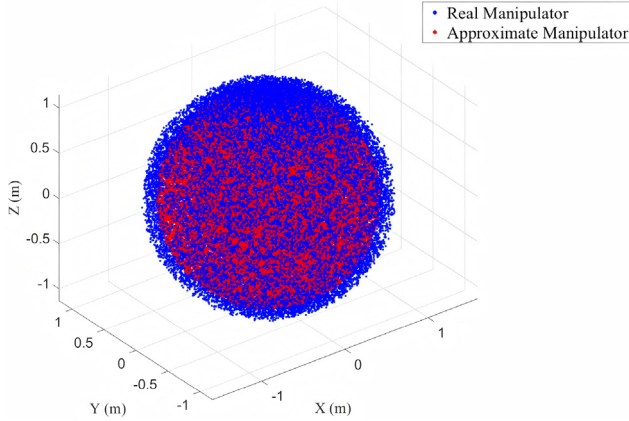


Fig. 9. Scatter plot of reachable points of the approximate manipulator obtained by traditional method and the real manipulator.

$$/L_i(\theta_{2(k+1)}, \theta_{6(k+1)}) = \theta_{i(k+1)} \propto (\theta_{2(k+1)}, \theta_{6(k+1)})(i = 1, 3, 4, 5, 7) \quad (49)$$

$$L_i(\theta_{2(k+1)}, \theta_{6(k+1)}) = \theta_{i(k+1)}(i = 2, 6) \quad (50)$$

These inequalities define the feasible region for θ_2 and θ_6 . After each iteration update, check if all constraints $L_i(\theta_{2(k+1)}, \theta_{6(k+1)})$ are satisfied. If any constraint is violated, project $x_{k+1} = \begin{pmatrix} \theta_{2(k+1)} \\ \theta_{6(k+1)} \end{pmatrix}$ back into the feasible region. This means adjusting x_{k+1} to satisfy $L_i(\theta_{2(k+1)}, \theta_{6(k+1)}) = 0$ for the violated constraints. This ensures all seven joint angles stay within their specified limits.

The initial guess significantly impacts the improved LM algorithm's solving success rate and efficiency. To address the algorithm's sensitivity to initial values and its tendency to fall into local optima, this paper proposes an approximate initial-value calculation method for SRS manipulators.

Current methods for approximating a 7-DOF manipulator with multiple offsets as an SRS manipulator typically involve zeroing the offsets to obtain an approximate SRS manipulator. However, for the manipulator studied here with offsets at the shoulder, elbow, and wrist, this approximation can place much of the real arm's workspace outside the approximate arm's workspace, as shown in Fig. 9. This discrepancy causes many real-arm postures to lie outside the approximate manipulator's workspace, making it impossible to solve for the approximate manipulator's joint angles for these postures and leading to initial-value solution failures. Consequently, the traditional method of obtaining an approximate manipulator needs optimizing. To make the workspace of the approximate manipulator include the workspace of the real manipulator as much as possible, we extended the approximate manipulator's d_3 and d_5 to $\sqrt{0.066^2 + 0.53^2} + 0.066$ as show in Fig. 10, enhancing its reach. The workspace comparison shows the approximate manipulator's workspace mostly covers the real manipulator's in Fig. 11. For the few real-

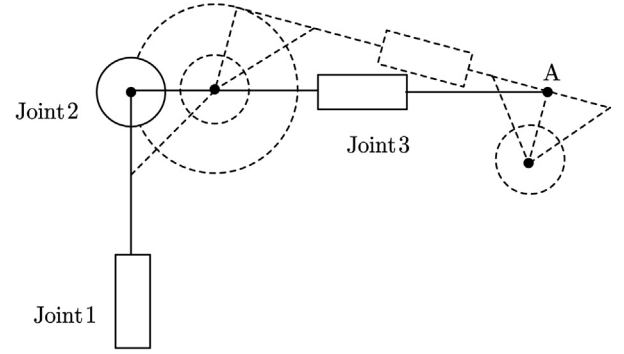


Fig. 10. The approximate SRS manipulator compared to the real manipulator.

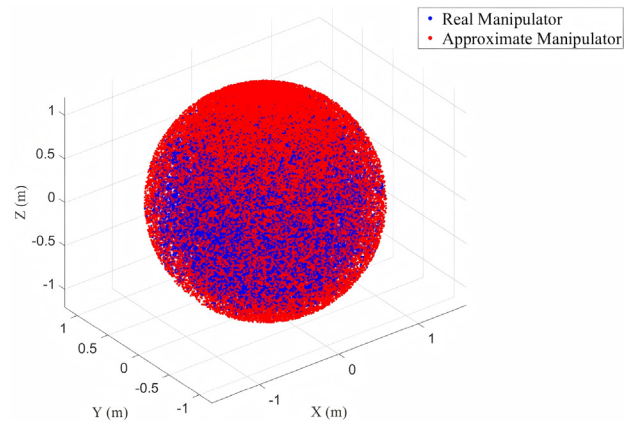


Fig. 11. Scatter plot of reachable points of the approximate manipulator obtained by the paper's method and the real manipulator.

manipulator postures outside the approximate manipulator's workspace, we set initial values to the middle of joint limits: $x_0 = \begin{pmatrix} \theta_{2(0)} \\ \theta_{6(0)} \end{pmatrix} = \begin{pmatrix} 63.75^\circ \\ 63.75^\circ \end{pmatrix}$. The SRS manipulator

inverse kinematics for each desired posture was solved using the MDH parameters in Table 3, and analytical solutions for the initial values $x_0 = \begin{pmatrix} \theta_{2(0)} \\ \theta_{6(0)} \end{pmatrix}$ were obtained.

These solutions, with minimal computational cost, significantly enhanced the improved LM algorithm's efficiency and speed. As the IK solution for the approximate SRS manipulator is analytical, computational load for obtaining initial values is minimal. Consequently, appropriate initial values greatly enhance the solving efficiency and speed of the improved LM algorithm.

4.2. The improved PSO algorithm

The improved initial-value calculation method for the improved LM algorithm boosts solving success rate and efficiency. To further enhance the algorithm, we adopted an improved PSO algorithm (Isiet and Gadala, 2020) for unsolved desired postures by the improved LM algorithm. This approach transforms the system of two nonlinear

Table 3
MDH parameters of the approximate SRS manipulator.

Joint	$a_{i-1}/(m)$	$\alpha_{i-1}/(^{\circ})$	$d_i/(m)$	$\theta_i/(^{\circ})$	Range/ $(^{\circ})$
1	0	0	0	$\theta_1 - 180$	$-170 \sim 170$
2	0	-90	0	$\theta_2 - 180$	$-42.5 \sim 170$
3	0	-90	$\sqrt{0.066^2 + 0.53^2} + 0.066$	$\theta_3 - 180$	$-170 \sim 170$
4	0.066	-90	0	$\theta_4 - 180$	$-42.5 \sim 170$
5	0.066	-90	$\sqrt{0.066^2 + 0.53^2} + 0.066$	$\theta_5 - 180$	$-170 \sim 170$
6	0	-90	0	$\theta_6 - 180$	$-42.5 \sim 170$
7	0	-90	0	$\theta_7 - 180$	$-170 \sim 170$

equations in two unknowns into an objective function minimization problem, leveraging the improved PSO algorithm to seek the global optimal solution within the given search space. It avoids reliance on initial values and effectively escapes local optima, ensuring high robustness.

For the system of two nonlinear equations in two unknowns, we first convert it into an objective function minimization problem. Then, the improved PSO algorithm is used to locate the global optimal solution within the specified search space. This method not only circumvents the dependence on initial values but also effectively escapes local optima, ensuring high robustness. Construct the objective function:

$$J(\theta_2, \theta_6) = F^2(\theta_2, \theta_6) + G^2(\theta_2, \theta_6) \quad (51)$$

When $J(\theta_2, \theta_6) = 0$, (θ_2, θ_6) is an exact solution to Eqs. (38) and (39). The problem then is reduced to finding solutions within joint limits that minimize $J(\theta_2, \theta_6)$ globally (ideally to 0).

This study designed the inertia weight using an exponential decay strategy. Compared to the conventional linearly decreasing method, exponential decay allows for more flexible inertia weight adjustment, effectively preventing premature convergence.

$$\omega_{new} = \omega_{min} + (\omega - \omega_{min}) \times \exp\left(-\frac{iter}{maxIterations}\right) \quad (52)$$

ω_{new} denotes the current iteration's inertia weight, with ω_{min} marking its minimum value. Here, ω refers to the inertia weight's initial state, $iter$ tracks the present iteration, and $maxIterations$ sets the iteration ceiling.

Particle distribution reflects swarm diversity and search status. A dynamic inertia weight adjustment mechanism based on particle distribution can better balance global exploration and local exploitation. The particle distribution index, measured by the standard deviation of particle positions, is calculated as follows:

$$distribution = \sqrt{\frac{1}{N \times D} \sum_{k=1}^N \sum_{j=1}^D (x_{k,j} - \bar{x}_j)^2} \quad (53)$$

N is the number of particles. D is the number of dimensions. $x_{k,j}$ is the position of particle k in dimension j . $\bar{x}_j$ is the average position of all particles in dimension j , calculated as follows:

$$\bar{x}_j = \frac{1}{N} \sum_{k=1}^N x_{k,j} \quad (54)$$

The inertia weight can be dynamically adjusted based on the calculated *distribution*. The specific adjustment rules are as follows:

$$\omega_{new} = \begin{cases} \max(\omega_{new}, \omega_{high}), & \text{if } distribution < threshold \\ \min(\omega_{new}, \omega_{low}), & \text{otherwise} \end{cases} \quad (55)$$

ω_{high} is the upper limit of the inertia weight when the distribution is small, used to enhance global exploration. ω_{low} is the lower limit of the inertia weight when the distribution is large, used to enhance local exploitation. *threshold* is the threshold of the distribution, used to determine the concentration of particle distribution.

The acceleration coefficients are designed using an exponential decay strategy. They decay from initial values c_1 and c_2 to minimum values c_{1min} and c_{2min} , balancing the impact of individual and group experience on particle velocity.

$$c_{1new} = c_{1min} + (c_1 - c_{1min}) \times \exp\left(-\frac{iter}{maxIterations}\right) \quad (56)$$

$$c_{2new} = c_{2min} + (c_2 - c_{2min}) \times \exp\left(-\frac{iter}{maxIterations}\right) \quad (57)$$

c_{1new} and c_{2new} are the acceleration coefficients at the current iteration. c_{1min} and c_{2min} are the minimum values of the acceleration coefficients. c_1 and c_2 are the initial acceleration coefficients. The particle velocity update formula integrates the adaptively tuned inertia weight and acceleration coefficients:

$$v_k(i+1) = \omega_{new} v_k(i) + c_{1new} r_1 (p_{kbest} - x_k(i)) + c_{2new} r_2 (g_{best} - x_k(i)) \quad (58)$$

$v_k(i+1)$ and $v_k(i)$ represent particle k 's velocity at iterations $i+1$ and i , respectively. p_{kbest} is particle k 's individual best position. g_{best} denotes the global best position. and r_1, r_2 are random values between 0 and 1.

$$x_k(i+1) = x_k(i) + v_k(i+1) \quad (59)$$

$x_k(i+1)$ and $x_k(i)$ denote the position of particle k at iterations $i+1$ and i , respectively.

To increase particle diversity and avoid premature convergence, a crossover strategy is adopted. The aim is to move particle positions towards g_{best} while retaining some of their current information. This helps prevent particles from gathering too soon around local optima, enhancing global exploration. By probabilistically adjusting particle positions, the search space coverage is increased, avoiding local optima traps. Specifically:

For each particle k , perform crossover with a probability ξ in each iteration. Generate a random number r in $[0,1]$ for each particle k .

If $r < \xi$, then perform the crossover operation:

$$x_k(i+1) = \xi x_k(i) + (1 - \xi)g_{best} \quad (60)$$

The crossover probability ξ ranges between $[0, 1]$.

This paper presents an improved PSO algorithm with further improvements. By incorporating dynamic parameter adjustment and probabilistic crossover mechanisms, the algorithm's performance and adaptability are significantly boosted.

As formulated in Eqs. (52)–(55), an adaptive inertia weight adjustment mechanism based on a particle distribution indicator is introduced. This mechanism dynamically balances global exploration and local exploitation according to the population diversity: when particles are widely dispersed, the inertia weight is automatically reduced to accelerate local convergence; when the distribution is concentrated, the inertia weight is increased to enhance global exploration and avoid premature convergence. Meanwhile, an exponential decay strategy is adopted for the acceleration coefficients (Eqs. (56)), enabling the algorithm to rely more on individual and social experience in the early search stage and focus on refined exploitation in the later stage. This strategy effectively reduces unnecessary iterations.

As expressed in Eqs. (61), a probabilistic crossover operation is incorporated. This strategy allows particles to learn from the global best solution with a certain probability while retaining part of their own information. This not only increases the population diversity, effectively preventing the algorithm from being trapped in local optima, but also reduces computational waste caused by frequent optimization restarts, thereby improving the overall solution efficiency from another perspective.

Multi-threaded parallel computing serves as a key engineering optimization in this work to address computational cost. Since the position update and objective function evaluation of each particle in the PSO are mutually independent, they offer inherent advantages for parallelization. We employ multi-threading to distribute the evaluation of the entire particle population across multiple CPU threads for synchronous execution. Compared to sequential computation, this parallelization significantly reduces the time required for N objective function evaluations that would otherwise be executed sequentially, theoretically approaching a linear speedup. This directly reduces the overall time per iteration, which is crucial for minimizing

the total time of the iterative process described by Eqs. (58) and (59).

Through the triple-layer optimization—comprising dynamic parameter adjustment, diversity preservation strategies, and underlying multi-threaded parallel computing—the improved PSO algorithm proposed in this paper achieves a substantial enhancement in computational efficiency while maintaining high precision and strong robustness.

4.3. The hybrid algorithm

Compared to the improved LM algorithm, the improved PSO algorithm enhances solution success rate but reduces solving efficiency. In this paper, the proposed hybrid algorithm not only achieves a higher success rate but also maintains high solving efficiency. The flowchart of the algorithm is shown in Fig. 12.

As mentioned earlier, we can solve for eight groups of solutions about $\theta_2, \theta_4, \theta_6$, thereby deriving solutions about $\theta_1, \theta_3, \theta_5$, and θ_7 . This allows us to determine the corre-

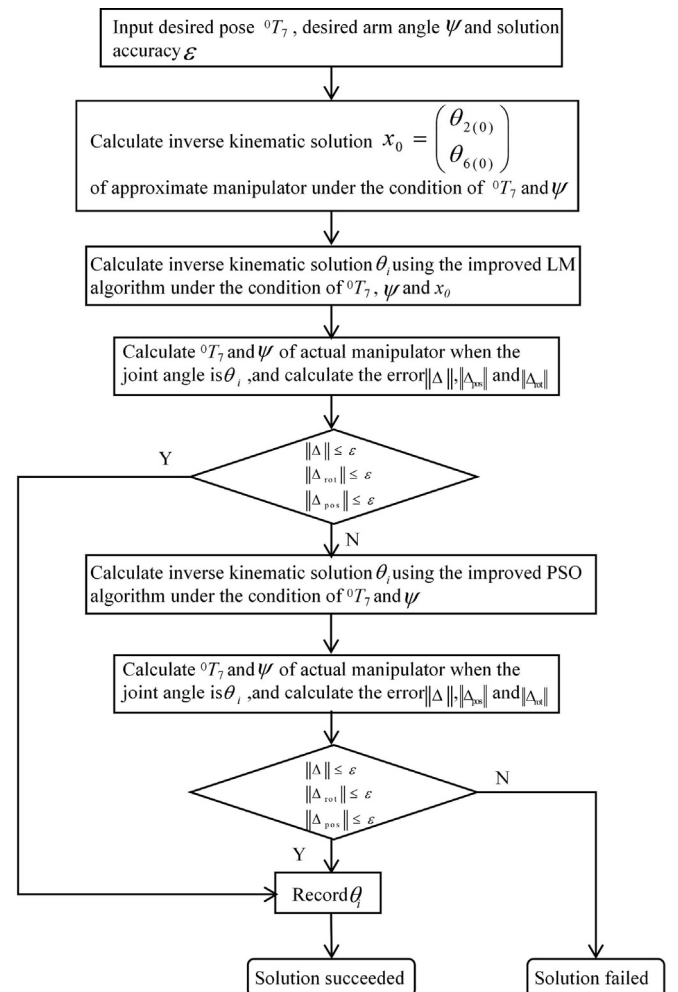


Fig. 12. The flowchart of the hybrid algorithm integrating the improved LM algorithm and the improved PSO algorithm.

sponding values for each group. Ultimately, we obtain eight sets of joint angles, from which the optimal set is chosen based on specific task requirements.

To validate the superiority of the LM-PSO algorithm, a detailed comparison was conducted with three hybrid methods: Levenberg–Marquardt–Genetic Algorithm (LM-GA), Levenberg–Marquardt–Simulated Annealing (LM-SA), and Levenberg–Marquardt–Ant Colony Optimization (LM-ACO). All algorithms were tested under the same experimental conditions, including the use of 10,000 randomly generated joint configurations based on the Monte Carlo method, and the same convergence accuracy requirements (position error $\leq 10^{-6}$ m, rotation error $\leq 10^{-6}$ rad). The experiments were performed on a platform equipped with an Intel(R) Core(TM) i9-14900KF processor and 64 GB of RAM to ensure a fair comparison. The results are presented in the Table 4. Data analysis indicates that the LM-PSO method significantly outperforms the other hybrid methods in terms of both success rate and computational efficiency. This advantage can be attributed to the following reasons: on the one hand, the PSO algorithm, owing to its social learning mechanism, exhibits higher convergence efficiency in global search compared to GA, SA, and ACO, enabling it to locate high-quality solution regions more rapidly; on the other hand, the LM algorithm possesses strong local optimization capabilities, allowing it to perform refined searches based on gradient information, thereby further enhancing the overall performance.

5. Experiments and comparative analysis

The inverse kinematics solution for the multi-offset 7-DOF manipulators based on the variable SRS equivalent manipulator, as proposed in Sections 3 and 4, was developed and executed in C++ using a computer featuring

an Intel(R) Core(TM) i9-14900KF 3.20 GHz processor and 64 GB RAM.

5.1. Inverse kinematics performance verification experiments

We specified 0T_7 in Eq. (61), $\psi = 150^\circ$, $\varepsilon = 10^{-6}$ m. Our proposed inverse kinematic algorithm yields 8 valid solutions. The arm angle error $\|\Delta\|$, position error $\|\Delta_{\text{pos}}\|$, and rotation error $\|\Delta_{\text{rot}}\|$ are presented in Table 5.

$${}^0T_7 = \begin{bmatrix} 0.8853 & -0.3510 & -0.3051 & -0.2311 \\ 0.3605 & 0.9324 & -0.0265 & 0.1014 \\ 0.2938 & -0.0866 & 0.9519 & 0.9596 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (61)$$

The proposed inverse kinematics method obtains 8 valid solution within joint limits, enabling multiple configuration options for diverse task requirements in manipulator operations. As shown in Table 5, the maximum arm angle error $\|\Delta\|$ is 3.2125×10^{-7} rad, and the minimum arm angle error is 3.6852×10^{-8} rad, which meets the precision requirements of actual manipulators. The maximum position error $\|\Delta_{\text{pos}}\|$ is 7.8636×10^{-8} m, with a minimum of 1.4505×10^{-8} m, while the maximum rotation error $\|\Delta_{\text{rot}}\|$ is 6.8173×10^{-7} rad and the minimum rotation error is 4.6739×10^{-7} m, ensuring compliance with on-orbit service task demands.

To evaluate the success rate and efficiency of the suggested technique, 5 experiments were conducted. Each experiment used the Monte Carlo method to generate sets of random joint angles within the joint limits of the multi-offset 7-DOF manipulator. These joint angles were used to compute desired poses 0T_7 and arm angles ψ . Inverse solutions were calculated under $\varepsilon = 10^{-6}$ m. The computation time per solution and success rates are illustrated in Fig. 13.

Table 4
Comparison between different hybrid algorithms.

Metric	THE PAPER	LM-GA	LM-SA	LM-ACO
Average Time/(ms)	0.6435	1.234	1.567	1.789
Solve Rate/(%)	98.434	92.1	90.5	91.8

Table 5
Inverse kinematics solution of the multi-offset 7-DOF robotic manipulator.

	$\theta_1/(^\circ)$	$\theta_2/(^\circ)$	$\theta_3/(^\circ)$	$\theta_4/(^\circ)$	$\theta_5/(^\circ)$	$\theta_6/(^\circ)$	$\theta_7/(^\circ)$	$\ \Delta\ /(\text{rad})$	$\ \Delta_{\text{pos}}\ /(\text{m})$	$\ \Delta_{\text{rot}}\ /(\text{rad})$
1	-99.7159	10.0895	46.8303	-39.3359	-19.4343	24.2775	92.4940	3.21×10^{-8}	7.86×10^{-8}	5.98×10^{-7}
2	-144.8340	7.3466	92.3082	-25.5634	154.1850	-17.9876	-81.4418	9.56×10^{-8}	2.93×10^{-8}	6.27×10^{-7}
3	66.9834	-8.3956	-120.0020	-33.3755	-22.3676	21.2596	95.5904	1.98×10^{-7}	5.13×10^{-8}	5.56×10^{-7}
4	18.8435	-7.6746	-71.5635	-20.7768	149.8910	-15.6950	-76.9289	3.69×10^{-8}	1.45×10^{-8}	6.82×10^{-7}
5	-69.2241	25.8935	-162.8330	75.0638	168.1050	41.5421	83.9921	1.61×10^{-7}	5.33×10^{-8}	6.82×10^{-7}
6	-82.7087	14.3204	-148.7450	51.8466	-15.0866	-30.7776	-93.3573	5.36×10^{-8}	1.83×10^{-8}	4.79×10^{-7}
7	106.7390	-20.5794	20.9159	62.0251	165.6250	34.2491	86.8443	1.46×10^{-7}	4.62×10^{-8}	4.67×10^{-7}
8	88.0309	-11.3125	40.1552	43.2709	-17.9287	-26.0802	-90.0961	5.19×10^{-8}	1.61×10^{-8}	5.16×10^{-7}

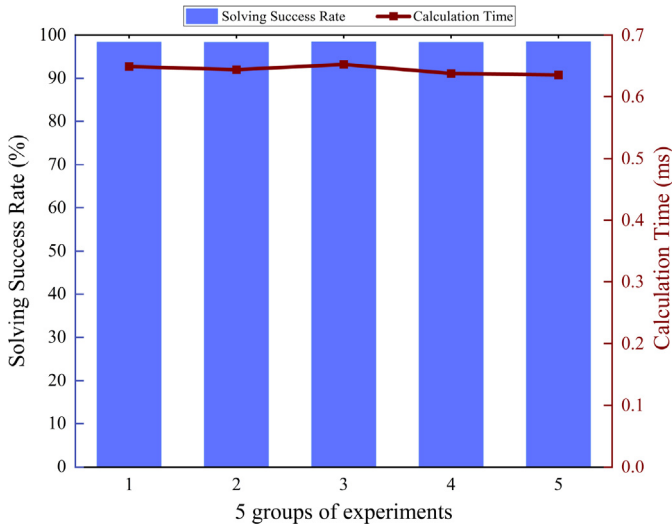


Fig. 13. The inverse solution performance.

As shown in Fig. 13, the 5 experimental groups have similar calculation times and success rates for individual solutions. The average calculation time per solution is 0.6435 ms, and the average success rate is 98.434 %. This indicates the suggested method has excellent computational effectiveness and consistent performance.

As shown in Table 6, the solution time increases as the required accuracy threshold decreases. For general tasks, a required accuracy of 10^{-6} m achieved a 98.43 % success rate, with an average computation time of 0.6788 ms per solution, meeting real-time control requirements. For high-precision tasks ($\varepsilon = 10^{-8}$ m), the success rate remains 97.96 % with an average computation time of 0.8853 ms. These results demonstrate that the proposed algorithm maintains high computational efficiency and success rates across varying precision requirements.

5.2. Application in trajectory planning of the multi-offset 7-DOF manipulators

To additionally validate the practicality of the proposed inverse kinematics method, it was employed in trajectory tracking for the multi-offset 7-DOF manipulator. A circular trajectory with a center coordinate of (0.6767, 0.2093, 0.0540) m, a radius of 0.3 m, and 200

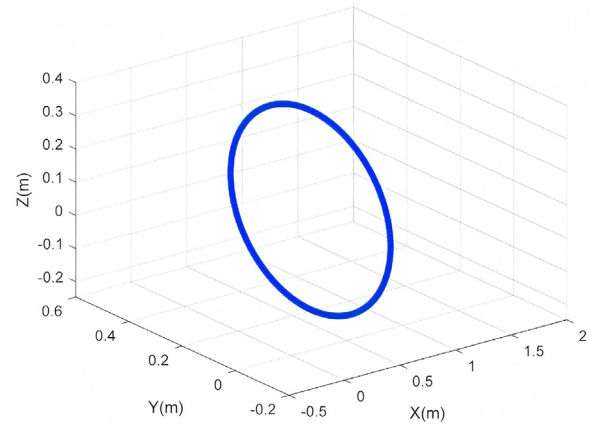


Fig. 14. Testing trajectory of the multi-offset 7-DOF manipulator.

equally spaced points in the Y-Z plane was generated, as illustrated in Fig. 14. The manipulator's initial joint angles were set to (64.1826°, 97.3564°, -128.6159°, 102.8932°, 30.2749°, 140.1928°, 119.6494°). The trajectory tracking was performed by maintaining a constant orientation matrix 0R_7 and fixed arm angle ψ , with $\psi = 60^\circ$ and $\varepsilon = 10^{-6}$ m predefined.

As presented in Fig. 15, the angular variations of each joint are plotted. The joint trajectories exhibit continuous and smooth variations, meeting the stringent continuity requirements for joint motion in aerospace applications. To ensure statistical validity, the algorithm was executed 100 times for repeatability analysis. The results demonstrate an average computation time of 0.5892 ms per trajectory point and a total path execution time of 1.1784 s, with a solution success rate of 100 %.

5.3. Comparison experiments and analysis

This section validates the proposed inverse kinematics method through comparative experiments. As no existing arm-angle-based IK methods are available for the studied multi-offset 7-DOF robotic arm configuration, we benchmark our approach against three general-purpose algorithms (KDL, TRAC-IK, Bio IK) and two variants of our method using improved LM (Chen and Ma, 2023) and improved PSO (Isiet and Gadala, 2020) solvers. All algorithms share identical precision settings ($\|\Delta_{\text{pos}}\| = 10^{-6}$ m, $\|\Delta_{\text{rot}}\| = 10^{-6}$ rad). Using 50,000 Monte

Table 6
The performance of the algorithm with different required accuracy.

Required accuracy/(m)	Single solution calculation time/(ms)	Solving success rate/(%)
10^{-4}	0.5521	99.01
10^{-5}	0.5697	98.94
10^{-6}	0.6435	98.43
10^{-7}	0.7241	98.02
10^{-8}	0.8853	97.96
10^{-9}	0.9212	97.20

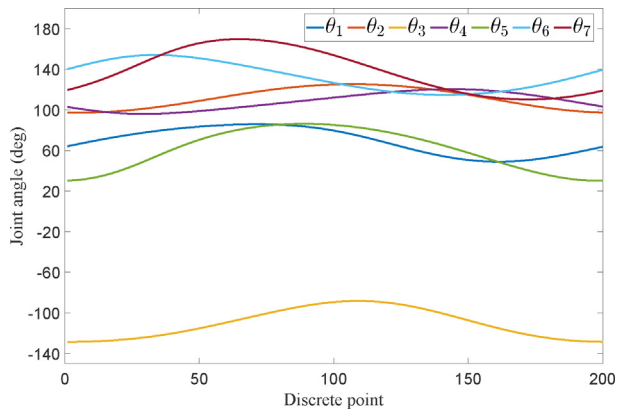


Fig. 15. Simulation results for the circular trajectory.

Carlo-generated joint configurations, Table 7 summarizes the average computation time and success rates of all six methods.

Table 7 shows the variable equivalent manipulator based method with the improved LM (Chen and Ma, 2023) has a low solution success rate (88.09 %) for 2D nonlinear equations. This is because the LM-based algorithm is sensitive to initial values and tends to fall into local optima. However, using the approximate-manipulator-based initial-value strategy increases the success rate to 93.75 % and reduces the single-solution time to 0.54 ms. This strategy works because the approximate and equivalent manipulator have similar joint angles in most postures, leading to initial values close to the real manipulator's joint angles, thus reducing iterations and avoiding local optima. Although solving the approximate manipulator's joint angles adds a step, it is an analytical solution with a short time (0.003 ms per initial value), much less than the time saved by the strategy (0.6 ms per solution). When using the improved PSO (Isiet and Gadala, 2020), the equivalent-manipulator-based method is slow (7.6042 ms per solution) due to PSO's high computational load. Though the improved PSO in this study reduces the time by 4.277 ms per solution, it still doesn't meet on-orbit service requirements. However, it offers a high success rate (95.72 %). The hybrid strategy, combining both methods, achieves a high success rate (98.434 %) and a reasonable time (0.6435 ms) by using the improved LM method with the initial-value strategy for most cases and the improved PSO for the remaining 6.25 % of failed cases.

Current hybrid algorithms for nonlinear equations often use intelligent algorithms (e.g., PSO) to find approximate

initial values and then use optimization algorithms (e.g., LM) to converge to precise solutions. But in this study, this approach is less effective than the proposed hybrid strategy. The improved LM method with the initial-value strategy already has a 93.75 % success rate, and the time to obtain analytical initial values is much shorter than that for intelligent algorithms to find approximate solutions. This phenomenon to some extent demonstrates the superiority of the method based on the concept of variable equivalent manipulator in this paper. Therefore, despite longer solving times for a few (4.684 %) postures, the proposed strategy has a better average solving time, meeting on-orbit service needs.

The KDL algorithm has a low inverse kinematics solving efficiency (50.86 %) for the manipulator in this study. In contrast, TRAC-IK and Bio IK have higher solution success rates (92.84 % and 96.25 %, respectively). Their average solving times per solution are 0.7529 ms, 0.8446 ms, and 0.8724 ms, respectively. The proposed method outperforms KDL, TRAC-IK, and Bio IK in solution success rate and efficiency and can easily introduce arm angles as redundant parameters.

In on-orbit servicing missions, motion stability of robotic systems is a critical performance metric, with joint motion continuity being a decisive factor. To evaluate this, we conducted trajectory-tracking experiments with the robotic end-effector following a circular path (radius: 0.3 m, 200 via-points) while the base remained fixed. Joint angle differentials between consecutive points were analyzed, yielding 199 differential sets per method. As shown in Fig. 16, conventional solvers exhibit hazardous discontinuities:

KDL exhibits hazardous jumps at the 129th group (Joints 1, 3, 5, and 7: 1.0656 rad, 1.3150 rad, 4.9372 rad, 1.2736 rad), 193rd group (Joints 5, 6, and 7: 3.1902 rad, 3.7118 rad, 3.2465 rad), and 194th group (Joints 5, 6, and 7: 3.1179 rad, 3.7150 rad, 3.1547 rad). TRAC-IK fails at the 21st group (Joints 2, 3, and 7: 1.8820 rad, 3.0828 rad, 2.4627 rad), while Bio IK diverges at the 141st group (Joints 1, 3, 6, and 7: 1.8258 rad, 4.1774 rad, 2.4997 rad, 3.2054 rad). These abrupt angular shifts (>1 rad) induce velocity spikes exceeding safety thresholds and pose collision risks. In contrast, our method based on variable equivalent manipulator maintains maximal differentials of 0.0202 rad (below the 0.05 rad safety threshold), ensuring smooth velocity profiles and configuration consistency for collision-free operations, thereby validating its superiority

Table 7

Average time and solve rate of different method.

Metric	THE PAPER	LM	PSO	KDL	TRAC-IK	Bio IK
Average Time/(ms)	0.6435	1.11923	7.6042	0.7529	0.8446	0.8724
Solve Rate/(%)	98.434	88.09	95.72	50.86	92.84	96.25
$\ \Delta_{\text{pos}}\ /(m)$	10^{-6}	10^{-6}	10^{-6}	10^{-6}	10^{-6}	10^{-6}
$\ \Delta_{\text{rot}}\ /(rad)$	10^{-6}	10^{-6}	10^{-6}	10^{-6}	10^{-6}	10^{-6}
$\ \Delta\ /(rad)$	10^{-6}	10^{-6}	10^{-6}	\	\	\

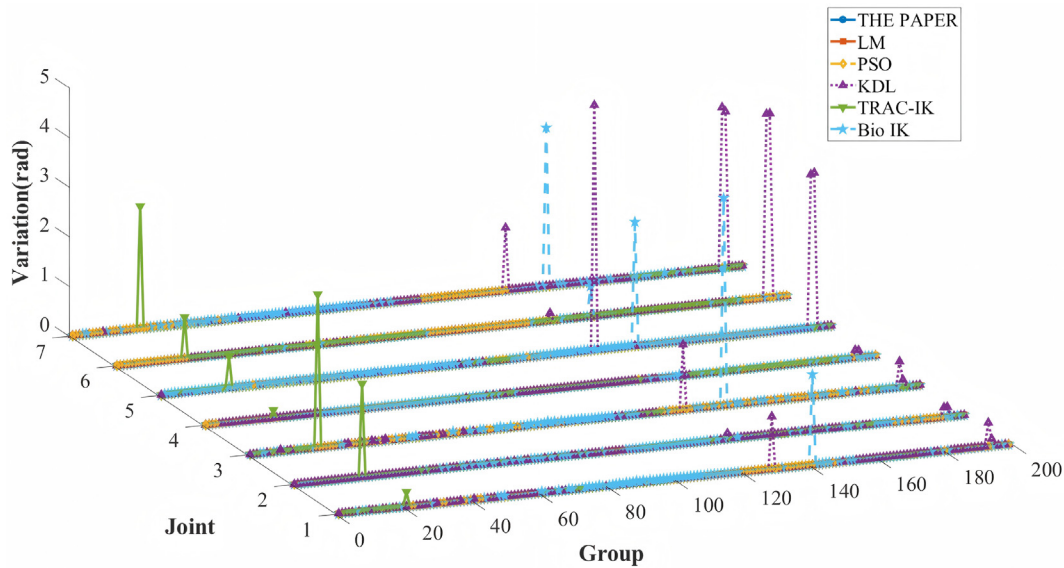


Fig. 16. The variation of joints during the continuous motion of the manipulator.

in securing mission-critical stability under extreme orbital constraints.

5.4. Discussion

On-orbit servicing tasks impose extremely stringent requirements on the performance of inverse kinematics (IK) algorithms, which are mainly reflected in three dimensions: solution stability, accuracy, and efficiency, and these directly determine the success or failure of the mission. The IK solver proposed in this paper, based on the variable SRS equivalent model and hybrid optimization algorithm, is designed precisely to meet these core constraints in the on-orbit environment. First, in terms of solution stability, the reliability of the algorithm is paramount. Traditional numerical IK methods (such as KDL) or intelligent algorithms (such as Bio IK), due to their sensitivity to initial values or inherent randomness, are prone to generating jumping solutions in joint space, causing joint torque impacts and threatening the safety of the spacecraft platform. As shown in the comparative experiments in Fig. 16, traditional methods exhibited mutations exceeding 1 radian in trajectory tracking, while our algorithm, through arm angle parameterization and equivalent modeling, transforms the problem into a deterministic search in low-dimensional space, ensuring continuous and smooth joint trajectories with a maximum differential of only 0.0202 radians, fundamentally guaranteeing motion stability.

In terms of solution accuracy, the algorithm achieves pose errors on the order of below 10^{-6} meters/radians, meeting the extreme requirements of on-orbit precision assembly. In terms of solution efficiency, the hybrid algorithm (LM-PSO) supplemented by an initial value strategy based on an approximate model achieves an average single solution time of 0.6435 ms and a success rate of 98.434 %, capable of meeting the computational resource constraints

of real-time on-orbit control. In summary, the excellent performance of the proposed inverse kinematics algorithm in stability, accuracy, and efficiency provides a solid technical foundation for addressing the challenges of on-orbit servicing.

Furthermore, the currently available inverse kinematics methods for the robotic manipulator configurations studied in this paper, such as KDL, Tracik, and BioIK, are all based on Jacobian pseudo-inverse iteration algorithms. A notable drawback of such algorithms is the randomness in their solutions. Even with the same parameter settings, solving the same trajectory path multiple times may yield different inverse kinematics solutions. This difference manifests in two forms. The first, as mentioned in Subsection 5.3, is that the obtained joint angles may exhibit significant jumps. The second refers to cases where the joint angles from multiple solves of the same trajectory are all continuous, yet differ across different solving instances. While this may be tolerable in certain ground-based scenarios, it is absolutely unacceptable in on-orbit environments. This uncertainty in the solution could potentially damage the spacecraft or the manipulator, and the cost of repair in orbit is prohibitively high. Consequently, such kinematics algorithms are not considered acceptable in practical engineering applications.

In contrast, the inverse kinematics algorithm proposed in this paper incorporates an arm-angle parameter constraint. Under identical configuration parameters, solving the same trajectory yields joint angles that are both continuous and identical. To better demonstrate the superior solving stability of our algorithm in on-orbit environments compared to existing methods, we designed a comparative experiment. Different algorithms were used to solve the circular trajectory described in Subsection 5.2 for 10,000 iterations, and their joint angle jumps and solution consistency were compared. The experimental results in Table 8 show

that our algorithm produced no joint angle jumps in all 10,000 solves, and all 10,000 results were identical. In contrast, KDL, Tracik, and BioIK exhibited frequency of joint discontinuity of 15.03 %, 9.74 %, and 8.21 %, respectively, and produced different results in 20.32 %, 10.63 %, and 3.82 % of the solves, respectively.

Unlike ground-based environments, on-orbit applications impose extremely high demands on the solving stability of inverse kinematics algorithms—a requirement that existing methods clearly fail to meet. Our proposed method satisfies this requirement, which constitutes one of the key contributions of this study.

5.5. Null space exploration

To evaluate the performance of null-space exploration, we fix the end-effector in a certain pose Eq. (62) and change the arm angle parameter ψ from 0° to 360° . The results are shown in Table 9, demonstrating that the proposed algorithmic framework can reliably output stable and accurate solutions for any given redundancy parameter input, thereby validating its generality and robustness.

$${}^0T_7 = \begin{bmatrix} -0.2037 & 0.8579 & 0.4716 & 0.4896 \\ -0.5652 & -0.4964 & 0.6589 & -0.5825 \\ 0.7994 & -0.1324 & 0.5860 & -0.5099 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (62)$$

In practical on-orbit servicing missions, ψ should not be selected arbitrarily. Rather, it serves as a critical “control handle” for achieving superior operational capabilities. The optimal selection of ψ is a typical constrained optimization problem, which can be mathematically formulated as finding an optimal ψ^* that extremizes an objective function $\Phi(\theta, \phi)$ based on the joint angles $\theta(\psi)$, while satisfying constraints such as joint limits. For instance, in the obstacle avoidance scenario, ψ may be

defined as the negative value of the minimum distance between the manipulator and surrounding obstacles, thereby maximizing this distance to avoid collisions. In the joint pose optimization scenario, ψ can be formulated as the sum of deviations of the joint angles from their initial positions, thus minimizing this deviation to maintain the manipulator in a dexterous configuration.

5.6. Workspace boundary and singularity analysis

To evaluate the performance of the algorithm near the workspace boundaries, 10,000 points near the edges of the workspace are randomly generated using the Monte Carlo method. These points are obtained by restricting the joint angles to values close to the allowable limits (e.g., $\theta_2 \approx 180^\circ$, $\theta_6 \approx 180^\circ$), and computing the corresponding end-effector poses. The experimental results demonstrate that the average success rate of the algorithm at the boundary points is 96.8 %, slightly lower than that at interior points (98.434 %), yet remains high. The average computation time is 0.712 ms, which is marginally longer than that for interior points (0.6435 ms), attributable to the increased likelihood of numerical instability near the workspace limits. These results confirm that the algorithm maintains strong robustness even at the workspace boundaries.

Regarding the issue of singularities, it should be emphasized that the proposed method—based on a reconfigurable equivalent model and arm angle parameterization—inherently avoids the traditional Jacobian matrix singularities. This can be explained as follows: Traditional Jacobian singularities arise from the geometric characteristics of the manipulator configuration (e.g., wrist singularity, elbow singularity). However, the proposed method transforms the inverse kinematics problem into solving two nonlinear equations via an equivalent model, thereby circumventing

Table 8
Comparison between different hybrid algorithms.

Metric	THE PAPER	KDL	TRAC-IK	Bio IK
Frequency of joint discontinuity/(%)	0	15.03	9.74	8.21
Frequency of solution inconsistency/(%)	0	20.32	10.63	3.82

Table 9
Null space exploration.

$\psi/(^\circ)$	$\theta_1/(^\circ)$	$\theta_2/(^\circ)$	$\theta_3/(^\circ)$	$\theta_4/(^\circ)$	$\theta_5/(^\circ)$	$\theta_6/(^\circ)$	$\theta_7/(^\circ)$
0	-55.5717	170.3674	0	-70.3081	124.7682	112.0195	102.8412
45	-11.4201	151.0598	86.0375	-65.4295	92.7734	141.3603	113.6198
90	-21.2189	118.9152	124.9032	-70.1440	12.2560	166.0013	79.5104
135	-38.9854	101.5746	147.8227	-64.6829	-58.0171	153.0800	40.8931
180	-58.5886	98.0786	-179.3005	-60.1588	-96.1621	130.8191	36.5633
225	-77.0559	107.3703	-144.5766	-58.4514	-129.5381	109.6741	44.8358
270	-89.6486	126.2477	-111.3193	-58.9697	-165.3391	95.6251	61.2105
315	-92.1791	150.1820	-74.4706	-62.4037	156.0833	94.6099	82.3525
360	-55.5717	170.3674	0	-70.3081	124.7682	112.0195	102.8412

the direct inversion of the Jacobian matrix. The introduction of the arm angle parameter ψ provides an additional degree of redundancy, enabling the avoidance of singular configurations near traditional singular regions. For instance, when the manipulator approaches wrist singularity, adjusting ψ can alter the elbow position, thus evading the singular configuration. Within the framework of the equivalent model, singularities manifest as ill-conditioned equations (e.g., denominators approaching zero). Nevertheless, the proposed hybrid algorithm (LM-PSO) effectively handles these scenarios through global search and local optimization, preventing solution failures.

5.7. Real world experiment

We conduct a polyline trajectory tracking experiment on the left arm of an ABB YuMi robot. The initial pose is specified in Eq. (63). Relative to the end-effector coordinate system, the end-effector first moves 0.2 m in the negative z-direction, followed by a 0.3 m movement in the positive x-direction. At the same time, the end posture remains unchanged and the arm angle remains unchanged at 30° ($\psi = 30^\circ$).

$${}^0T_7 = \begin{bmatrix} 0.8853 & -0.3510 & -0.3051 & -0.2311 \\ 0.3605 & 0.9324 & -0.0265 & 0.1014 \\ 0.2938 & -0.0866 & 0.9519 & 0.9596 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (63)$$

The experimental setup is shown in Fig. 17, which is consistent with the simulation conditions. The joint angles and end-effector trajectories are shown in Fig. 18, demonstrating that our algorithm also delivers excellent performance in a real-world system. It is particularly worth noting that during actual operation, the robot arm exhibited smooth and continuous joint motion without any sudden jumps

or oscillations, which is fully consistent with the stability advantages observed in simulations.

5.8. Experimental validation on other manipulator configurations

To demonstrate the applicability of the proposed method to other 7-DOF manipulators with multiple offsets, the inverse kinematics algorithm derived using our approach was tested on the Franka Research 3 (FR3) manipulator. The kinematic diagram of the FR3 manipulator is shown in Fig. 19, and its MDH parameters are provided in Table 10. It can be observed that the FR3 is a 7-DOF manipulator with a distinct geometric configuration, and its MDH parameters differ significantly from those of the MIRROR/YuMi-like structure primarily studied in this work.

Based on the proposed concept of variable SRS equivalence, the FR3 manipulator was analyzed using the same

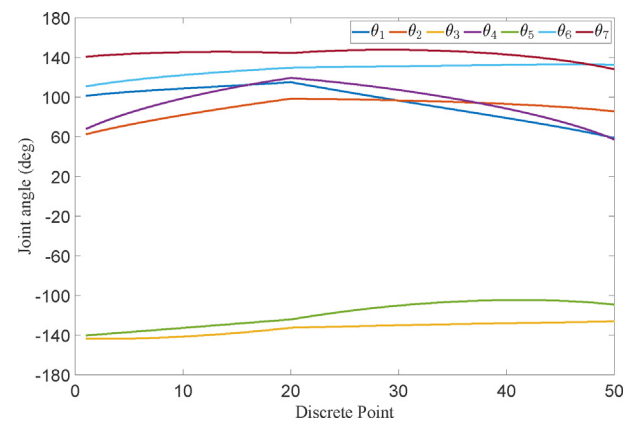


Fig. 18. Physical experiment of polyline trajectory joint angle curve.

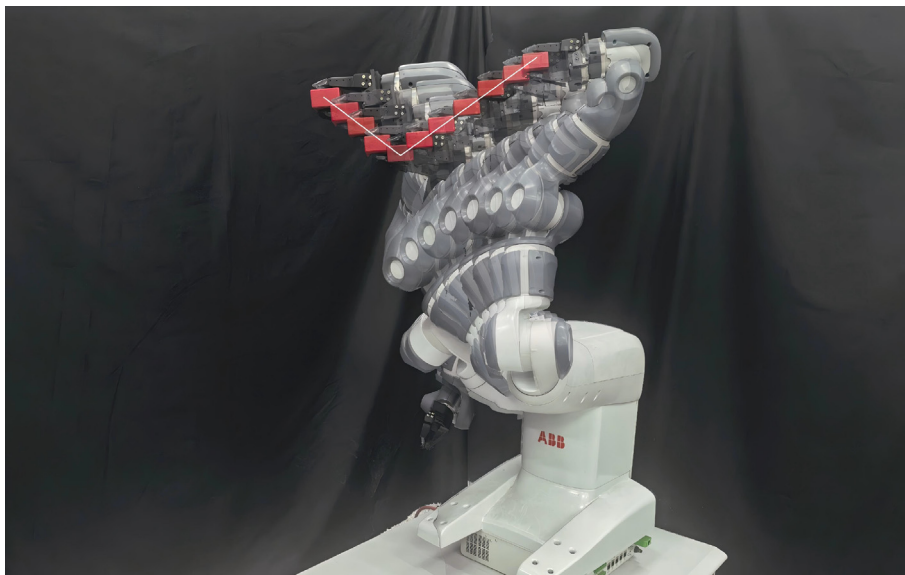


Fig. 17. Real world experiment.

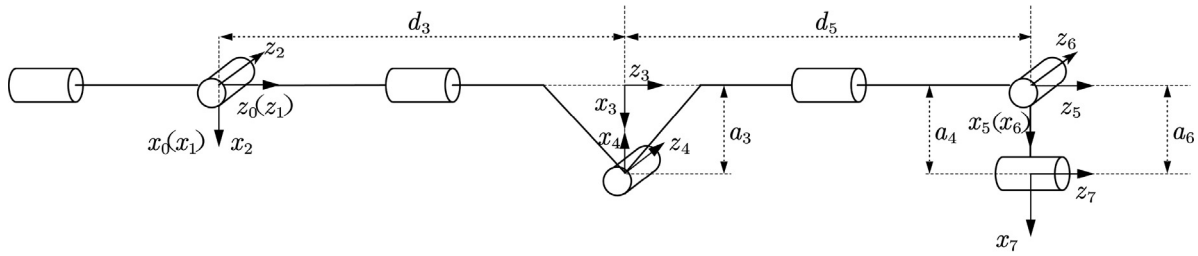


Fig. 19. The kinematic model of the FR3.

methodology outlined in this paper. Fig. 20 illustrates the definition of the shoulder and wrist points for the FR3 and provides a schematic of its variable equivalent manipulator. Fig. 21 shows the coordinate system diagram of the variable equivalent manipulator, where the dashed circles represent the joint positions of the actual manipulator, and the solid circles denote those of the equivalent manipulator. The corresponding MDH parameters for the variable equivalent manipulator of the FR3 are listed in Table 11.

Following the subsequent steps of the proposed algorithm, the inverse kinematics solution for the FR3 manipulator was derived and its effectiveness was evaluated. Given a desired pose as expressed in Eq. (64) and $\psi = 36^\circ$, the obtained inverse kinematics solutions and their corresponding accuracy are presented in Table 12, where $\|\Delta\|$ denotes the arm angle error, $\|\Delta_{\text{pos}}\|$ represents the position error, and $\|\Delta_{\text{rot}}\|$ indicates the orientation error. The results demonstrate that the proposed method maintains its effectiveness, high accuracy, and arm angle controllability even when applied to the FR3 manipulator, which exhibits a significantly different configuration from the manipulator primarily investigated in this study.

$${}^0T_7 = \begin{bmatrix} -0.7878 & 0.5101 & -0.3453 & -0.0243 \\ -0.5548 & -0.8311 & 0.0381 & 0.2600 \\ -0.2675 & 0.2216 & 0.9377 & 0.5052 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (64)$$

It should be noted that the solutions for Groups 5 to 8 yield negative values for θ_4 , which fall outside the joint limit of Joint 4 defined for the FR3 manipulator. This occurs because four of the eight equations derived from the arm-

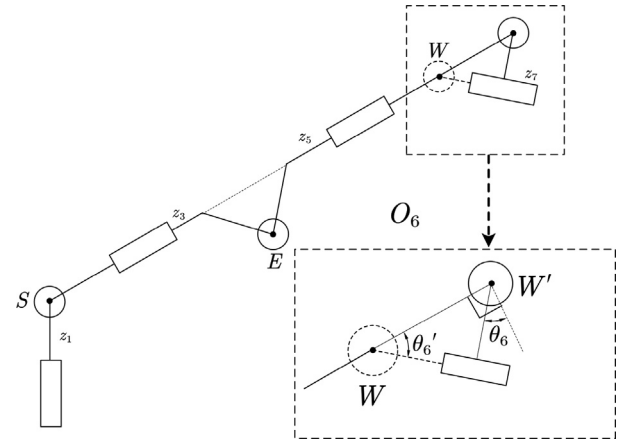


Fig. 20. Definition of shoulder and wrist points of the FR3 and schematic of variable equivalent manipulator.

angle-based algorithm search for solutions in the negative domain of θ_4 , as indicated in Eq. (17) and Eq. (18) of this paper. Other studies offer parallel evidence. Zhao et al. (2023), for example, demonstrated a similar pathway through compelling experiments. It is important to emphasize that the joint limits given in this study are specific to the FR3 product. For other homologous manipulators that permit negative values for Joint 4, a complete set of eight valid solutions within the joint limits can still be obtained.

Additionally, experiments were conducted to evaluate the solution success rate and computational efficiency. Using the Monte Carlo method, 10,000 sets of joint angles were randomly generated within the joint limits of the 7-DOF manipulator with multiple offsets to compute the

Table 10
MDH parameters of the FR3.

Joint	$a_{i-1}/(m)$	$\alpha_{i-1}/(^{\circ})$	$d_i/(m)$	$\theta_i/(^{\circ})$	Range/ $(^{\circ})$
1	0	0	0	θ_1	$-132 \sim 132$
2	0	-90	0	θ_2	$-87 \sim 87$
3	0	90	0.316	θ_3	$-143 \sim 143$
4	0.0825	-90	0	$\theta_4 + 180$	$26 \sim 157$
5	0.0825	-90	0.384	$\theta_5 + 180$	$-142 \sim 142$
6	0	-90	0	θ_6	$-132 \sim 61$
7	0.088	90	0	θ_7	$-154 \sim 154$

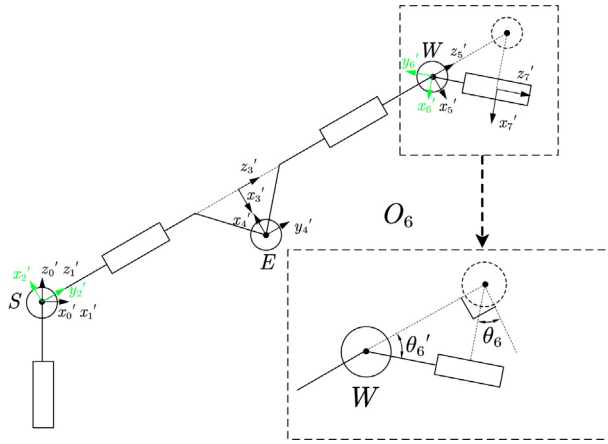


Fig. 21. Coordinate system diagram for the variable equivalent manipulator of the FR3.

desired pose and arm angle. The inverse kinematics solutions were calculated under a given solution accuracy. The average computation time per solution and the success rate were 0.62 ms and 99.01 %, respectively. The experimental results demonstrate that the proposed algorithm is applicable to the FR3-type 7-DOF manipulator, thereby confirming its potential for application to a wider range of 7-DOF manipulator configurations with multiple offsets.

6. Conclusion and future work

This work proposes a novel inverse kinematics method specifically designed for the multi-offset 7-DOF manipulators. To address the failure of traditional shoulder and wrist point definitions in such manipulators due to non-

coplanar joint axes, a revised definition method is introduced. The concept of variable SRS equivalent manipulator is proposed, resolving the inability of existing methods to incorporate the arm angle as a redundant parameter. By utilizing the variable equivalent manipulator framework, the complex problem of solving seven joint angles under joint limits is reduced to solving two nonlinear equations with two unknowns within feasible domains. For the nonlinear system, a hybrid algorithm integrating an improved LM method and an improved PSO is developed, coupled with an effective initial-value estimation strategy. This significantly improves solution success rates (98.434%) and computational efficiency (0.6435 ms). The proposed method fills the research gap in applying arm angle-based solutions to the studied manipulator configuration. Compared to general-purpose algorithms (KDL, TRAC-IK, Bio IK), this method demonstrates superior success rates, efficiency, and stability, providing a safer and more reliable IK solution for on-orbit servicing manipulator studied in this paper.

Our work is tailored for the on-orbit environments due to the following two reasons. Firstly, our method features high real-time performance and high reliability. On-orbit servicing missions impose extremely high demands on computational efficiency and determinism, and the design of our inverse kinematics algorithm has consistently targeted these objectives. Our key innovation lies in the introduction of a "variable equivalent SRS arm" concept, which transforms the complex inverse kinematics problem of a 7-DOF manipulator into solving a low-dimensional (two-variable) system of nonlinear equations. This dimensional-ity reduction is inherently aimed at improving computa-

Table 11
MDH parameters for the variable equivalent manipulator of the FR3.

Joint	$a_{i-1}/(m)$	$\alpha_{i-1}/(^{\circ})$	$d_i/(m)$	$\theta_i/(^{\circ})$	Range/ $(^{\circ})$
1	0	0	0	θ_1	$-132 \sim 132$
2	0	-90	0	θ_2	$-87 \sim 87$
3	0	90	0.316	θ_3	$-143 \sim 143$
4	0.0825	-90	0	$\theta_4 + 180$	$26 \sim 157$
5	0.0825	-90	$0.384 - (0.088/\sin\theta_6)$	$\theta_5 + 180$	$-142 \sim 142$
6	0	-90	0	θ_6	$-132 \sim 61$
7	0	90	$0.088/\tan\theta_6$	θ_7	$-154 \sim 154$

Table 12
Inverse kinematics solution of the FR3.

	$\theta_1/(^{\circ})$	$\theta_2/(^{\circ})$	$\theta_3/(^{\circ})$	$\theta_4/(^{\circ})$	$\theta_5/(^{\circ})$	$\theta_6/(^{\circ})$	$\theta_7/(^{\circ})$	$\ \Delta\ /(rad)$	$\ \Delta_{pos}\ /(m)$	$\ \Delta_{rot}\ /(rad)$
1	134.7643	87.0416	-165.8810	105.7082	39.2717	-43.6308	-138.9071	5.27×10^{-8}	3.28×10^{-8}	9.22×10^{-7}
2	116.2587	71.7359	-161.7831	98.5592	-136.6509	51.6320	54.4809	7.68×10^{-8}	1.29×10^{-8}	2.37×10^{-7}
3	-45.2357	-87.0416	14.1190	105.7082	39.2717	-43.6308	-138.907	1.99×10^{-8}	3.15×10^{-8}	5.56×10^{-7}
4	-63.7413	-71.7359	18.2169	98.5592	-136.6509	51.6320	54.4809	2.88×10^{-8}	1.45×10^{-8}	3.93×10^{-7}
5	131.4342	57.4972	15.8449	-56.0861	-125.1114	-29.5140	-163.7629	2.22×10^{-8}	6.95×10^{-8}	3.66×10^{-7}
6	98.6405	46.6734	26.2606	-37.8340	78.9619	35.9091	13.2765	2.42×10^{-8}	3.32×10^{-8}	6.93×10^{-7}
7	-48.5658	-57.4972	-164.1551	-56.0861	-125.1114	-29.5140	-163.7629	2.33×10^{-8}	7.97×10^{-8}	1.00×10^{-7}
8	-81.3595	-46.6734	-153.7394	-37.8340	78.9619	35.9091	13.2765	9.33×10^{-8}	2.55×10^{-8}	3.70×10^{-7}

tional efficiency. The proposed hybrid algorithm (improved LM and improved PSO) is specifically devised to address the computational constraints in on-orbit systems. The improved LM algorithm achieves high-precision solutions (position error $\leq 10^{-6}$ m) with high efficiency in most cases (average computation time: 0.6435 ms per solution). For the few complex configurations, the improved PSO algorithm ensures avoidance of local minima, thereby guaranteeing strong robustness and a high success rate (98.434 %). This combination of high efficiency and high reliability is essential for meeting the real-time control demands in on-orbit operations. Secondly, our method ensures the motion stability. Traditional numerical iterative methods (e.g., KDL) or intelligent algorithms (e.g., Bio-IK), when solving such complex configurations, often suffer from discontinuous jumps in solutions due to initial value sensitivity or randomness, as shown in Fig. 16. These jumps can directly cause sudden changes in joint velocities or torques, posing significant risks to the spacecraft platform. In contrast, our method parameterizes the redundancy via the arm angle parameter, enabling systematic and continuous generation of eight deterministic solutions for a single end-effector pose. As illustrated in Fig. 15, this allows for the selection of the smoothest and most stable solution from the set, thereby providing a solid foundation at the kinematic level for high-level trajectory planning and vibration control. This capability is crucial for enhancing on-orbit operational performance.

This paper addresses the inverse kinematics problem of a 7-DOF manipulator with multiple offsets for on-orbit servicing by proposing a novel method based on a variable SRS-equivalent manipulator and hybrid optimization. The approach systematically resolves the core challenge that traditional arm-angle-based methods cannot be applied to such complex manipulator configurations. The main contributions of this work are summarized as follows:

First, at the theoretical level, a new concept of a "variable SRS-equivalent manipulator" is proposed. This concept serves as the theoretical foundation of this study. By establishing an equivalent model that dynamically adapts to the end-effector pose, it effectively overcomes the bottleneck of traditional shoulder/wrist point definition methods, thereby enabling the powerful tool of arm-angle parameterization to be applied to the manipulator configuration under investigation.

Second, at the methodological level, an efficient dimensionality-reduction solving framework is constructed. By transforming the problem of solving seven joint angles into solving a system of two nonlinear equations, the proposed method fundamentally reduces computational complexity. Validation and comparative experiments demonstrate that, compared to general-purpose algorithms such as KDL, TRAC-IK, and Bio-IK, the proposed method exhibits significant advantages in solution success rate (98.434 %), computational efficiency (average 0.6435 ms per solution), and motion stabil-

ity, meeting the stringent real-time and reliability requirements of on-orbit tasks.

Finally, at the algorithmic level, a hybrid improved optimization strategy is designed. This strategy deeply integrates the advantages of an improved Levenberg–Marquardt algorithm and an improved Particle Swarm Optimization algorithm, supplemented by an efficient initial value estimation strategy, thereby equipping the solver with both global search capability and rapid local convergence. Experimental results confirm that the algorithm effectively handles the solution of the nonlinear system of equations, providing robust computational support for the implementation of the proposed method.

Future work will investigate its applicability to other manipulator configurations and explore its integration with dynamic obstacle avoidance and dynamics control. Building upon the high-performance method developed in this study, we will proceed to design controllers that account for on-orbit dynamic characteristics, aiming to ultimately fulfill the comprehensive requirements of complex on-orbit servicing missions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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